Endmember Extraction Guided Transferable Adversarial Attack for Hyperspectral Image Classification

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***Abstract*—In recent years, adversarial attacks in hyperspectral image (HSI) classification have attracted increasing attention. However, most existing approaches rely on the loss or gradient information of a single classifier to optimize pixel-level adversarial perturbations, which undermines the reflectance consistency of hyperspectral data and limits the transferability of adversarial examples across different classifier architectures. These drawbacks compromise both the physical interpretability and the practical applicability of adversarial examples. To address these challenges, this article proposes an endmember extraction guided transferable adversarial attack (EEGTAA) framework for generating adversarial examples that preserve reflectance consistency while achieving the strong transferability. Specifically, EEGTAA first employs endmember extraction techniques to obtain the endmember composition of HSIs and introduces adversarial perturbations into the endmember matrix, thereby ensuring the physical plausibility of the generated samples. Subsequently, EEGTAA integrates the loss gradients from multiple heterogeneous classifiers to optimize perturbations on the endmember matrix, enhancing the threat to target classifiers with unknown architectures and parameters. Furthermore, a conflict-mitigation gradient optimization strategy is theoretically derived, which utilizes weighted loss gradients to correct the perturbation update direction within the hyperspherical neighborhood of the averaged gradient, alleviating the suboptimal optimization caused by divergent gradient directions across classifiers. Experimental results on benchmark HSI datasets demonstrate that EEGTAA not only produces adversarial examples with high-attack performance and reflectance consistency but also exhibits superior transferability across classifier architectures.**

***Index Terms*—Adversarial attack, classification, endmember extraction, hyperspectral images (HSIs).**

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I. Introduction

HYPERSPECTRAL images (HSIs), acquired by imaging spectrometers, constitute a 3-D data cube that integrates both spatial and spectral dimensions, with spectral resolution reaching the nanometer level [1]. Unlike traditional RGB or multispectral images, each pixel in an HSI corresponds to a continuous spectral curve that captures reflectance or emission information across hundreds of contiguous spectral bands. Such rich and detailed spectral information provides higher precision and reliability in characterizing the chemical composition and physical structure of materials [2], [3], [4]. As one of the core tasks in HSI interpretation, the HSI classification aims to assign category labels to individual pixels, playing a pivotal role in a wide range of applications, including Earth observation, military reconnaissance, and environmental monitoring [5], [6], [7].

In earlier studies, researchers employed traditional machine learning approaches, such as extreme learning machines [8], support vector machines [9], and rotation forests [10], [11], for HSI classification. However, these methods generally rely heavily on manually designed feature engineering, which requires substantial domain expertise and intricate parameter tuning. Moreover, they are often tailored to specific tasks, resulting in high complexity as well as limited generalization and robustness [12], [13]. In recent years, deep neural networks (DNNs), with their superior feature extraction capability and joint spectral–spatial modeling, have significantly improved the accuracy and efficiency of HSI classification [12], [14], [15], [16], [17], [18], [19]. Zhang et al. [12] applied convolutional neural network to HSI classification, where stacked 2-D convolutional layers were used to extract discriminative features and spatial structural information of pixels. To more comprehensively capture the spectral–spatial characteristics of hyperspectral data, a spectral–spatial visual attention mechanism was incorporated into the convolutional feature extractor, thereby enhancing the network's sensitivity to the most relevant and informative features [13]. Recognizing that 2-D convolutional layers often neglect interband correlations, Ghaderizadeh et al. [14] introduced 3-D convolutional layers into the classifier and employed optimization techniques, such as batch normalization and exponentially decaying learning rates to alleviate overfitting. ResMorCNN further utilized four morphological operators to extract shape and structural details of images, thereby achieving more refined classification

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results, particularly in edge regions [15]. In addition, several researchers have explored the use of transformer architectures to model long-range dependencies across spectral bands and extract global features, achieving remarkable improvements in classification performance [16], [17], [18].

Although the aforementioned DNNs have demonstrated outstanding performance in HSI classification, their robustness remains limited, as they are typically effective only on clean, unperturbed data [20]. Existing studies have shown that when imperceptible adversarial perturbations are added to benign input examples, DNNs may output incorrect predictions—a phenomenon known as adversarial attacks. Goodfellow et al. [21] first introduced the idea of generating adversarial examples by applying a single-step gradient of the loss function with respect to the input, thereby posing a direct threat to DNNs. To further strengthen the attack capability, subsequent works developed iterative gradient-based methods or employed generative models to synthesize adversarial perturbations [22], [23], [24], [25], [26]. For HSIs, the high-dimensional data structure offers greater flexibility and a larger attack space, which motivated recent studies to design attack strategies specifically tailored to mislead hyperspectral classifiers [27], [28], [29], [30], [31], [32]. For example, Burnel et al. [27] proposed an improved Wasserstein generative adversarial network (GAN) that reweights the real data distribution to guide the generator in producing adversarial examples. However, training such generators often requires significant computational resources and is prone to mode collapse. To address this, an improved DeepFool method was proposed under the assumption of access to the target model, which generates adversarial examples near the decision boundary by applying minimal perturbations to hyperspectral inputs [28]. Nevertheless, these approaches generally overlook the robustness of generated adversarial pixels against image-processing-based defense mechanisms. To mitigate this limitation, some studies introduced explicit modeling of local smoothness in adversarial perturbations or optimized perturbations in the latent space, thereby producing hyperspectral adversarial examples that are more resilient to spectral and spatial filtering [29], [30].

However, the aforementioned adversarial attack methods often lack a thorough understanding of the inherent physical properties of hyperspectral data. Due to the limited spatial resolution of hyperspectral imaging sensors, the spectral response of a single pixel is typically a mixture of multiple pure material spectra, where the pure reflectance spectra are referred to as endmembers, and their corresponding proportions are known as abundances [33]. This phenomenon is commonly referred to as the *reflectance consistency* of HSIs, where the spectral of each pixel can be interpreted as a combination of several endmembers with specific proportions. As illustrated in Fig. 1, existing adversarial attack methods for HSI classification typically generate adversarial samples threatening to a single victim model by tampering pixel values. However, such approaches not only overlook the intrinsic physical properties of HSIs but also exhibit limited practicality in real-world applications. On the one hand, pixel-level perturbations fail to account for the reflectance consistency of HSIs and may result in adversarial pixels with abrupt spectral distortions.

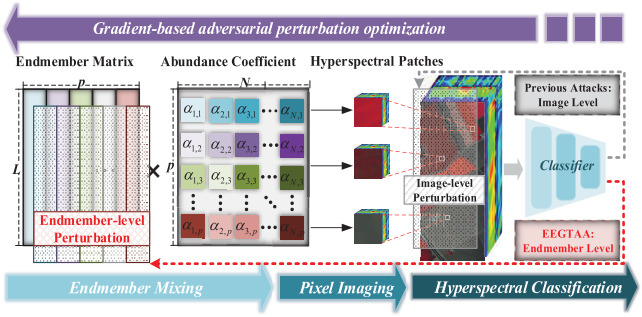


Fig. 1. Illustration of the HSI imaging chain and the differences between EEGTAA and previous attack methods.

These anomalous spectra violate natural physical laws and can thus be easily detected by physical defense mechanisms. On the other hand, in real-world scenarios, attackers often cannot access the architecture and parameters of the target HSI classifiers. When adversarial examples are optimized solely on a single-surrogate model, the optimization process is prone to poor local optima, which weakens their transferability across different model architectures and consequently reduces their practical threat [23], [24].

To address the aforementioned limitations, this study proposes an endmember extraction guided transferable adversarial attack (EEGTAA) framework for HSI classification, aiming to generate adversarial examples that pose threats across models with different architectures. Specifically, EEGTAA first analyzes the endmember composition of each pixel in HSI using endmember extraction techniques and estimates the corresponding abundances via fully constrained least squares [34], [35], [36]. Considering the imperceptibility and nonnegativity requirements of hyperspectral adversarial examples, a constrained optimization model is then formulated to optimize perturbations applied at the endmember level, where the perturbed endmembers guide the generation of adversarial samples. Subsequently, EEGTAA leverages the ensemble classification loss of multiple hyperspectral classifiers with diverse architectures to guide the optimization of adversarial perturbations through backpropagation, thereby enhancing their transferability across models. Finally, a conflict-mitigation gradient ascent strategy is theoretically derived, in which weighted loss gradients are employed to calibrate the perturbation update direction within the hyperspherical neighborhood of the average gradient, thereby alleviating suboptimal optimization caused by conflicting loss gradient directions across hyperspectral classifiers with different architectures. Since perturbations are applied at the endmember level rather than by directly altering pixel values, the generated adversarial examples exhibit improved spectral smoothness and demonstrate stronger robustness against image preprocessing-based defense mechanisms. In summary, the main contributions of this study can be outlined as follows.

1) This study is the first to incorporate the reflectance consistency into adversarial attacks on HSIs, where adversarial perturbations are applied at the endmember

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level to guide the generation of adversarial examples, rather than directly altering pixel spectral.

2) This study initially proposes an ensemble-based adversarial attack method for HSI classification, in which the optimization of perturbations is guided by backpropagating the ensemble loss of multiple hyperspectral classifiers, while simultaneously satisfying imperceptibility and nonnegativity constraints.
3) This study designs a conflict-mitigation gradient optimization strategy, where the weighted loss gradients are employed to correct the perturbation update direction within the neighborhood of the average gradient, thereby mitigating the suboptimal optimization caused by conflicting gradient directions across different classifiers.

The remainder of this article is organized as follows. Section II reviews related work on adversarial attacks for HSI and endmember extraction. Section III presents and analyzes the proposed EEGTAA method. Section IV describes the datasets, experimental settings, and results. Section V concludes this article and provides further discussion.

II. Background

This section provides a brief review of related studies on adversarial attacks for HSIs, as well as the fundamental concepts and key techniques of hyperspectral endmember extraction.

A. Adversarial Attacks for HSIs

Artificial intelligence methods represented by DNNs have significantly advanced image recognition, object detection, and natural language processing. However, DNNs are highly vulnerable to adversarial attacks and lack sufficient adversarial robustness [37], [38]. Goodfellow et al. [21] first introduced a single-step method that perturbs the input $\boldsymbol{x}$ with the gradient of the loss function with respect to the input, thereby misleading the model to make incorrect predictions

$$\boldsymbol{x}^{adv} = \boldsymbol{x} + \epsilon \cdot sign\left(\nabla_{\boldsymbol{x}} J\left(\boldsymbol{x}, y; \boldsymbol{\theta}\right)\right). \tag{1}$$

Nevertheless, single-step perturbations usually exhibit limited attack strength. Subsequent studies demonstrated that adversarial examples can be made more powerful by iteratively updating the perturbation with a smaller step size α [39]

$$\begin{aligned} \boldsymbol{x}_{t+1}^{adv} &= \boldsymbol{x}_t^{adv} + \alpha \cdot sign\left(\nabla_{\boldsymbol{x}} J\left(\boldsymbol{x}_t^{adv}, y; \boldsymbol{\theta}\right)\right) \\ \boldsymbol{x}_0^{adv} &= \boldsymbol{x}. \end{aligned} \tag{2}$$

To further enhance the attack performance against unseen target models, more recent research has focused on approaches, such as generative architectures, gradient editing, and semantic similarity, aiming to produce adversarial examples with high transferability [40], [41], [42], [43].

To date, extensive studies have been conducted on adversarial examples in the RGB domain, whereas research in the HSI domain remains relatively scarce. Compared with conventional RGB image classification tasks, HSI classification faces unique challenges, such as limited training data and higher susceptibility to model overfitting, making adversarial examples potentially more detrimental in this context [44]. Shi et al. [29] investigated the high-dimensional redundancy of HSI and the vulnerability of pixelwise perturbations to defense strategies, and proposed a universal object-level attack. Their method enforces local similarity of perturbations by constructing a spectrally superpixelized template with local smoothness constraints, thereby improving the robustness of adversarial examples against image-processing-based defenses. To further improve stealthiness, an improved DeepFool method was introduced, which jointly exploits loss and gradient information of the hyperspectral classifier to iteratively optimize adversarial examples [28]. This approach generates samples near the decision boundary with minimal perturbation magnitude. In addition, an optimized one-pixel attack strategy, OPAS, was developed, leveraging an improved differential evolution algorithm to design perturbations, while employing Latin hypercube initialization to better accommodate the high-dimensional characteristics of HSI, thereby improving attack efficiency [31]. Moreover, GANs have also been employed to craft HSI adversarial examples. For instance, Burnel et al. [27] modified the loss function of Wasserstein GAN and reweighted the real data distribution, guiding the generator to produce the adversarial spectral that are more natural and consistent with real spectral distributions.

B. Hyperspectral Endmember Extraction

The endmember extraction is a critical step in HSI interpretation, aiming to identify the pure spectral signatures (endmembers) that constitute a scene from complex mixed pixels. The accuracy of endmember extraction directly influences subsequent applications, such as feature classification, target detection, and environmental monitoring [45]. The linear mixture model (LMM), serving as the theoretical foundation for the endmember extraction, assumes that the spectral response of each mixed pixel can be expressed as a linear combination of multiple endmember spectra, thereby providing a computationally tractable framework for endmember extraction. If hyperspectral pixel $\boldsymbol{p} \in \mathbb{R}^{L\times 1}$ consists of p endmembers, the linear spectral mixing model can be expressed as follows:

$$\boldsymbol{p} = \sum_{i=1}^{p} a_i \boldsymbol{e}_i + \boldsymbol{\varepsilon}, \quad \text{s.t. } a_i \geq 0 \text{ and } \sum_{i=1}^{p} a_i = 1 \tag{3}$$

where a_i is the proportion (abundance) of the ith endmember $\boldsymbol{e}_i$ in $\boldsymbol{p}$ and $\boldsymbol{\varepsilon}$ is the model error term.

In recent years, endmember extraction algorithms based on LMMs have emerged, and many scholars have proposed corresponding endmember extraction algorithms. Winter [35] proposed the N-FINDR algorithm, which argues that in a high-dimensional space, a minimal convex packet containing all data points can be defined by p vertices, and the pure endmembers are located in the vertex positions of the convex packet. The N-FINDR algorithm first randomly selects p points as initial endmembers, calculates the corresponding monomorphic volume, iterates over all image elements, and finally arrives at a set of image elements corresponding to the endmembers when the maximum volume is reached. The endmember matrix $\boldsymbol{E} \in \mathbb{R}^{L\times p}$ needs to be augmented

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with a row of all-1 vectors to compute the volume. Let the endmember spectral be $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_p$, then the augmentation matrix is

$$\boldsymbol{E}_A = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \boldsymbol{e}_1 & \boldsymbol{e}_2 & \cdots & \boldsymbol{e}_p \end{bmatrix}. \tag{4}$$

The volume of the simplex is proportional to the absolute value of the determinant of the augmentation matrix

$$V(\boldsymbol{E}) = \frac{1}{(p-1)!} abs\left(|\boldsymbol{E}_A|\right) \tag{5}$$

where $\boldsymbol{e}_i$ denotes the spectral vector of the ith endmember. The primary objective of the N-FINDR algorithm is to identify a set of endmember combinations $\boldsymbol{E}$ that maximizes the corresponding hypervolume $V(\boldsymbol{E})$.

Although the N-FINDR algorithm is simple and easy to implement, it is computationally intensive, slow, and the results are unstable. The vertex component analysis (VCA) algorithm is more suitable for real hyperspectral datasets, as it offers lower computational complexity and higher unmixing accuracy compared with the N-FINDR algorithm [34]. The VCA algorithm begins by applying dimensionality reduction to the hyperspectral data $\boldsymbol{X} \in \mathbb{R}^{L \times N}$, where N denotes the number of pixels. The data are projected into an m-dimensional subspace, yielding the matrix $\boldsymbol{Y} \in \mathbb{R}^{m \times N}$. This step not only removes part of the noise but also reduces computational complexity. Subsequently, the reduced data matrix $\boldsymbol{Y}$ is column-normalized to facilitate the subsequent projection operations, with the normalized data denoted by $\boldsymbol{Y}'$. VCA algorithm then iteratively estimates the endmember matrix $\boldsymbol{E}$. For the $(k+1)$th iteration, in order to identify a new direction orthogonal to the k already extracted endmembers, an orthogonal projection operator $\boldsymbol{P}$ is constructed. If the subspace spanned by the current endmembers is denoted as $\langle \boldsymbol{E}_k \rangle$, the projection operator onto its orthogonal complement is defined as follows:

$$\boldsymbol{P} = \boldsymbol{I} - \boldsymbol{E}_k \left(\boldsymbol{E}_k^T \boldsymbol{E}_k\right)^{-1} \boldsymbol{E}_k^T \tag{6}$$

where $\boldsymbol{I}$ is the identity matrix. This projection ensures that the new direction remains independent of the previously identified endmembers, thereby enabling the extraction of an additional endmember. Subsequently, the normalized data matrix $\boldsymbol{Y}'$ is projected onto a test direction determined by the projection operator $\boldsymbol{P}$ and a randomly selected vector $\boldsymbol{u} \in \mathbb{R}^{m \times 1}$, yielding

$$\boldsymbol{v} = \boldsymbol{P}\boldsymbol{u}, \quad \boldsymbol{w} = \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|}. \tag{7}$$

For this direction $\boldsymbol{w}$, the projection values of all pixels are computed as follows:

$$f\left(\boldsymbol{y}'\right) = \boldsymbol{w}^T \boldsymbol{y}', \quad \boldsymbol{y}' \in \boldsymbol{Y}' \tag{8}$$

and the pixel index corresponding to the maximum projection value is identified

$$i = \arg\max_j |f\left(\boldsymbol{y}'_j\right)|. \tag{9}$$

This pixel represents the candidate endmember extracted in the current iteration, with its extreme point corresponding to one vertex of the simplex along the given direction. The identified endmember vector $\boldsymbol{y}'_i$ is then appended to the endmember matrix $\boldsymbol{E}_k$, forming an updated matrix $\boldsymbol{E}_{k+1}$. By repeating this process p times, the complete set of p endmembers is ultimately obtained.

LMM-based endmember extraction algorithms include statistical methods, sparse regression methods, and artificial intelligence methods, in addition to geometric methods. Statistical methods are used to identify and extract endmembers in HSIs by introducing geometric constraints and statistical analysis. Typical methods include GLNMF [46] and DECA [47]. Sparse regression methods formulate spectral unmixing as a linear sparse regression problem, where a laboratory-derived spectral library serves as the reference. By imposing sparse constraints, these methods automatically identify the most representative pure endmember spectra from the library. Representative approaches include SUnSAL [48], CLSUnSAL [49], MOSU [50], and EMTU-WNS [51]. In contrast, artificial intelligence methods reformulate endmember extraction as a combinatorial or sequential optimization problem and leverage intelligent algorithms to efficiently solve these problems, thereby improving both accuracy and robustness of endmember extraction [52], [53], [54].

III. Methodology

This section first describes the optimization objective of EEGTAA for HSI classification. Subsequently, an ensemble-based adversarial attack approach is introduced to enhance the transferability of adversarial examples across different classifiers. To further improve the attack performance, a conflict-mitigation perturbation optimization strategy is developed to mitigate gradient conflict among classifiers. Finally, a comprehensive summary and analysis of the proposed EEGTAA framework is provided. The overall workflow and algorithm details of EEGTAA are shown in Fig. 2 and Algorithm 1.

A. Endmember Extraction Guided Adversarial Attack

Unlike conventional RGB images with three channels, each pixel in an HSI is typically composed of a mixture of reflectance spectra from multiple endmembers, where each endmember corresponds to the purest spectral signature of a ground object. Existing adversarial attack methods for hyperspectral classification generally introduce perturbations directly in the original pixel space to mislead classifiers. However, such approaches may compromise the reflectance consistency of hyperspectral pixels, resulting in adversarial samples that cannot be reconstructed as a linear mixture of endmembers, i.e., $\boldsymbol{p}_{adv} = \boldsymbol{A}\boldsymbol{u}$, thereby diminishing their physical interpretability. To address this issue, this article propose an EEGTAA framework for hyperspectral classification. This framework generates hyperspectral adversarial examples by applying perturbations at the endmember level, ensuring the preservation of reflectance consistency. Since endmembers intrinsically represent the fundamental reflectance properties of materials in HSIs, performing perturbations on them naturally aligns with the physical mechanism of spectral mixing, thereby enhancing both the realism and physical interpretability of the adversarial examples.

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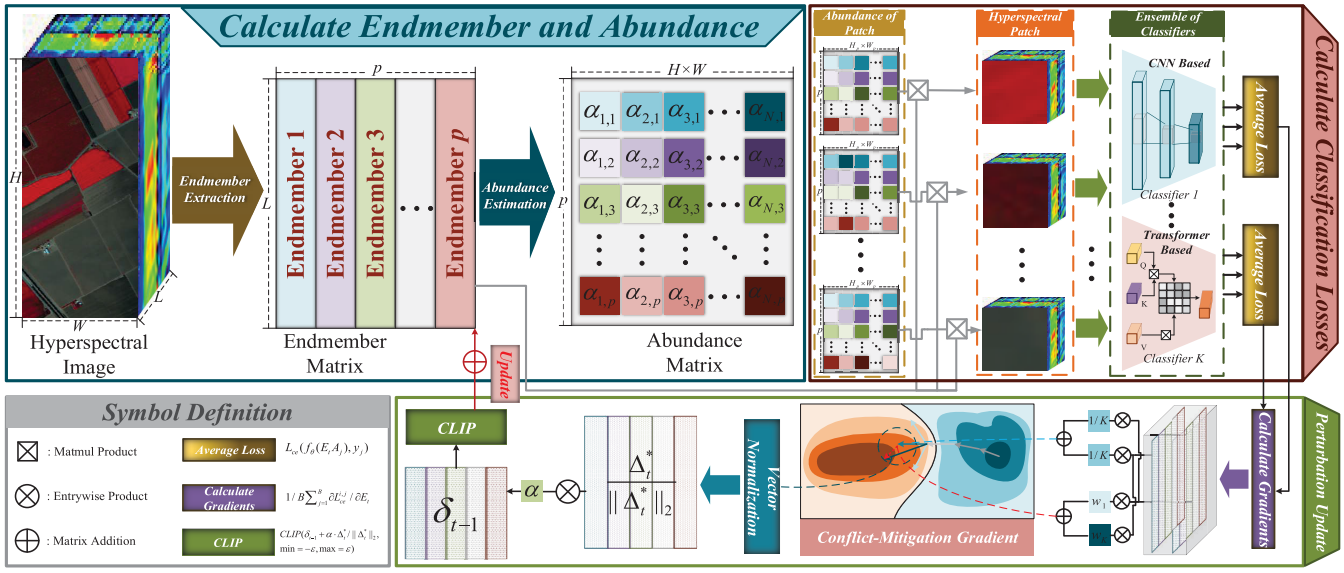


Fig. 2. Proposed EEGTAA framework consists of three stages. First, the blue component denotes the extraction of endmembers from the HSI using the VCA algorithm and the estimation of the corresponding abundance matrix. Second, the red component indicates feeding a mini-batch of hyperspectral patches into multiple HSI classifiers with different architectures to compute the corresponding classification losses. Finally, the green component represents backpropagating the gradients of each classifier with respect to the input and updating the adversarial perturbations on the endmember matrix through a conflict-mitigation gradient optimization strategy.

Algorithm 1 Algorithm of the EEGTAA

Input: A hyperspectral image $\boldsymbol{I}$; the ground-truth label y; the hyperspectral classifiers $f_\theta^1, f_\theta^2, \ldots, f_\theta^K$; the size of perturbation ϵ; the maximum number of iterations T_{max}; the step size α, the convergence hyperparameter c and the size of mini-batch B.

Output: Adversarial endmember matrix $\boldsymbol{E}_{adv}$ and hyperspectral adversarial samples $\boldsymbol{x}$.

***# Extract the Endmembers of HSI*;**
Extract endmember matrix $\boldsymbol{E}$ by VCA method.
Obtain the abundance vector $\boldsymbol{a}$ corresponding to each pixel $\boldsymbol{p}$ by solving Eq. (10).
***# Update the Adversarial Perturbation*;**
Initialize $\boldsymbol{\delta}_0 = \boldsymbol{0}$, $\boldsymbol{E}_0 = \boldsymbol{E}$.
for $t = 1 \rightarrow T_{max}$ **do**
    Update endmember matrix by $\boldsymbol{E}_t = \boldsymbol{E}_{t-1} + \boldsymbol{\delta}_{t-1}$.
    Construct a mini-batch $[\boldsymbol{x}_t^1, \boldsymbol{x}_t^2, \ldots \boldsymbol{x}_t^B]$ by randomly sampling abundance matrix and calculate hyperspectral adversarial samples by $\boldsymbol{x}_t^i = \boldsymbol{E}_t \boldsymbol{A}_i$.
    Update $\boldsymbol{\delta}_t$ by **Algorithm 2**.
end for
***# Output the Adversarial Endmember Matirix*;**
Calculate adversarial endmember matrix by $\boldsymbol{E}_{adv} = \boldsymbol{E} + \boldsymbol{\delta}_{T_{max}}$.
***# Output the Hyperspectral Adversarial Samples*;**
Calculate hyperspectral adversarial samples by $\boldsymbol{x}_{adv}^i = \boldsymbol{E}_{adv} \boldsymbol{A}_i$ *for* $i = 1, 2, \ldots, N$.
return $\boldsymbol{E}_{adv}$, $[\boldsymbol{x}_{adv}^1, \boldsymbol{x}_{adv}^2, \ldots, \boldsymbol{x}_{adv}^N]$.

For an HSI $\boldsymbol{I} \in \mathbb{R}^{L \times H \times W}$, the corresponding endmember matrix $\boldsymbol{E} \in \mathbb{R}^{L \times p}$ can be derived using the VCA algorithm [34], where $N = H \times W$ denotes the number of pixels in the HSI, L represents the number of spectral bands, and p is the number of extracted endmembers. Typically, each pixel $\boldsymbol{p} \in \mathbb{R}^{L \times 1}$ in $\boldsymbol{I}$ can be modeled as a linear combination of all endmembers weighted by their corresponding abundance coefficients $\boldsymbol{a} \in \mathbb{R}^{p \times 1}$, i.e., $\boldsymbol{p} = \boldsymbol{E}\boldsymbol{a}$. The abundance vector $\boldsymbol{a}$ can be estimated by solving the following optimization problem:

$$\arg\min_{\boldsymbol{a}} \|\boldsymbol{p} - \boldsymbol{E}\boldsymbol{a}\|_2^2 \quad \text{s.t. } a_i \geq 0, \ \sum_{i=1}^{p} a_i = 1 \tag{10}$$

where $a_i \geq 0$ and $\sum_{i=1}^{p} a_i = 1$ are the abundance nonnegativity constraint and abundance sum-to-one constraint, respectively [48].

EEGTAA introduces adversarial perturbations directly on the endmember matrix to guide the generation of adversarial hyperspectral pixels. The optimization objective is formulated as follows:

$$\begin{aligned} \boldsymbol{\delta} = \arg\max_{\boldsymbol{\delta}'} & \mathbb{E}_{\boldsymbol{a} \sim D(\boldsymbol{a})} \left[\mathcal{L}_{ce} \left(f_\theta \left(\left(\boldsymbol{E} + \boldsymbol{\delta}' \right) \boldsymbol{a} \right), y \right) \right] \\ & \text{s.t. } \|\boldsymbol{\delta}'\|_\infty \leq \epsilon, \ \boldsymbol{E} + \boldsymbol{\delta}' \geq 0 \end{aligned} \tag{11}$$

where f_θ denotes the HSI classifier parameterized by θ, $\boldsymbol{a}$ is the abundance vector corresponding to the hyperspectral pixel $\boldsymbol{p}$, and $D(\boldsymbol{a})$ represents the distribution of abundance vectors associated with labeled pixels in the HSI, y denotes the ground-truth label of pixel $\boldsymbol{p}$, $\mathcal{L}_{ce}(\cdot)$ is the classification loss defined by the cross-entropy function and $\boldsymbol{\delta}$ is the adversarial perturbation added to the endmember matrix $\boldsymbol{E}$. The constraint $\|\boldsymbol{\delta}'\|_\infty \leq \epsilon$ is introduced to limit the magnitude of the adversarial perturbation, thereby ensuring the imperceptibility of the generated hyperspectral adversarial pixels. The constraint $\boldsymbol{E} + \boldsymbol{\delta}' \geq 0$ enforces nonnegativity of all reflectance spectra in the perturbed endmember matrix, preserving the physical plausibility of the endmember representations.

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B. Ensembel-Based Adversarial Perturbation Generation

As shown in (11), this study aims to optimize an adversarial endmember matrix $\boldsymbol{E}_{adv}$, which is then used to generate adversarial hyperspectral pixel $\boldsymbol{p}$ through the formulation $\boldsymbol{p}_{adv} = \boldsymbol{E}_{adv}\boldsymbol{a}$, where $\boldsymbol{E}_{adv} = \boldsymbol{E} + \boldsymbol{\delta}$. Consistent with the existing gradient-based adversarial example generation methods [21], [39], the proposed approach leverages the gradient of the classification loss $\mathcal{L}_{ce}$ with respect to the endmember matrix $\boldsymbol{E}$ to effectively guide the optimization of the adversarial perturbation $\boldsymbol{\delta}$. Iterative optimization is commonly employed to enhance the effectiveness of adversarial attacks. Given the adversarial perturbation $\boldsymbol{\delta}_{t-1}$ obtained at iteration $t-1$ and $\boldsymbol{E}_t = \text{CLIP}(\boldsymbol{E}_{t-1} + \boldsymbol{\delta}_{t-1}, min = 0, max = 1)$, the gradient of the cross-entropy loss $\mathcal{L}_{ce}$ with respect to the endmember matrix $\boldsymbol{E}_t$ can be derived using the chain rule as follows:

$$\frac{\partial \mathcal{L}_{ce}}{\partial \boldsymbol{E}_t} = \frac{\partial \mathcal{L}_{ce}}{\partial \boldsymbol{p}_t} \cdot \frac{\partial \boldsymbol{p}_t}{\partial \boldsymbol{E}_t} = \frac{\partial \mathcal{L}_{ce}}{\partial \boldsymbol{p}_t} \cdot \boldsymbol{a}^T \tag{12}$$

where $\boldsymbol{p}_t = \boldsymbol{E}_t \cdot \boldsymbol{a}$ and $\text{CLIP}(\boldsymbol{x}, min = a, max = b)$ denote a truncation function that constrains the input $\boldsymbol{x}$ within the closed interval $[a, b]$. Traditional adversarial attack methods for hyperspectral classification typically optimize adversarial perturbation $\boldsymbol{\delta}$ by computing the gradient of the loss with respect to the pixel $\partial \mathcal{L}_{ce}/\partial \boldsymbol{p}_t$. However, such approaches often fail to preserve the reflectance consistency of adversarial pixels. In contrast, as illustrated in (12), EEGTAA introduces a perturbation term $\partial \boldsymbol{p}_t/\partial \boldsymbol{E}_t$ to propagate gradient information from the pixel level back to the endmember matrix, thereby enabling its targeted optimization. Moreover, the differentiable mapping between the pixel $\boldsymbol{p}_t$ and the endmember matrix $\boldsymbol{E}_t$ (e.g., $\boldsymbol{p}_t = \boldsymbol{E}_t\boldsymbol{a}$) ensures the smooth backpropagation of gradients.

Contemporary HSI classifiers typically enhance the classification of a central pixel by incorporating spectral information from its spatial neighbors, forming an input patch. To ensure the effectiveness of the generated adversarial examples against such classifiers, EEGTAA constructs a patch $\boldsymbol{x}$ comprising the central pixel and its adjacent pixels, which is then fed into the classifier. The resulting gradients with respect to the endmember matrix are subsequently used to guide the optimization of adversarial perturbations. Moreover, classifiers with different architectural designs leverage spectral features in distinct ways; for instance, CNN-based models exploit local spectral correlations, whereas transformer-based models capture global contextual dependencies across the patch. In practical scenarios, attackers typically do not have access to information such as the architecture, parameters, and gradients of the black-box target HSI classifier. They can only assess the success of the attack by querying the target classifier's predicted class outputs. To enhance the effectiveness of the adversarial endmember matrix across diverse classifier architectures and to ensure its threat potential against black-box target models with unknown structures, EEGTAA adopts an ensemble-based optimization strategy for adversarial perturbations and formulates the following optimization objective:

$$\begin{aligned} \boldsymbol{\delta} = \arg\max_{\boldsymbol{\delta}'} & \mathbb{E}_{f\sim\mathcal{F}}\left[\mathbb{E}_{\boldsymbol{A}\sim D(\boldsymbol{A})}\left[\mathcal{L}_{ce}\left(f_\theta\left(\left(\boldsymbol{E}+\boldsymbol{\delta}'\right)\boldsymbol{A}\right), y\right)\right]\right] \\ \text{s.t.} & \|\boldsymbol{\delta}'\|_\infty \le \epsilon,\ \boldsymbol{E}+\boldsymbol{\delta}' \ge 0 \end{aligned} \tag{13}$$

where $\boldsymbol{A} \in \mathbb{R}^{p\times H_p \times W_p}$ denotes the abundance matrix corresponding to all pixels within the input patch, and the patch fed into the classifier is defined as $\boldsymbol{x} = \boldsymbol{E}\boldsymbol{A}$, where H_p and W_p represent the height and width of the patch, respectively. The expectation term can be further expanded into the following summation form:

$$\begin{aligned} &\mathbb{E}_{f\sim\mathcal{F}}\left[\mathbb{E}_{\boldsymbol{A}\sim D(\boldsymbol{A})}\left[\mathcal{L}_{ce}\left(f_\theta\left(\left(\boldsymbol{E}+\boldsymbol{\delta}'\right)\boldsymbol{A}\right), y\right)\right]\right] \\ &= \frac{1}{KN}\sum_{i=1}^{K}\sum_{j=1}^{N}\mathcal{L}_{ce}\left(f_\theta^i\left(\left(\boldsymbol{E}+\boldsymbol{\delta}'\right)\boldsymbol{A}_j\right), y_j\right) \end{aligned} \tag{14}$$

where $\boldsymbol{A}_j$ and y_j denote the abundance matrix and ground-truth label of the jth patch $\boldsymbol{x}_j$, respectively, while f_θ^i represents the ith HSI classifier. It is important to note that the HSI classifiers used to optimize the adversarial endmember matrix are white-box for the attacker. By integrating the loss and gradient information from multiple white-box surrogate classifiers, the optimized adversarial endmember matrix becomes more threatening, effectively attacking black-box target classifiers.

To guide the adversarial optimization of the endmember matrix using the classification loss, the following computation is performed to estimate the corresponding optimization direction:

$$\begin{aligned} &\mathbb{E}_{f\sim\mathcal{F}}\left[\mathbb{E}_{\boldsymbol{A}\sim D(\boldsymbol{A})}\left[\frac{\partial \mathcal{L}_{ce}\left(f_\theta\left(\left(\boldsymbol{E}+\boldsymbol{\delta}'\right)\boldsymbol{A}\right), y\right)}{\partial \boldsymbol{\delta}}\right]\right] \\ &= \frac{1}{KN}\sum_{i=1}^{K}\sum_{j=1}^{N}\frac{\partial \mathcal{L}_{ce}\left(f_\theta^i\left(\left(\boldsymbol{E}+\boldsymbol{\delta}'\right)\boldsymbol{A}_j\right), y_j\right)}{\partial \boldsymbol{\delta}}. \end{aligned} \tag{15}$$

Computing the gradient of the endmember matrix based on the loss over all samples typically incurs substantial computational overhead. To address this, EEGTAA estimates the partial derivatives using mini-batches of hyperspectral patches during optimization

$$\begin{aligned} &\mathbb{E}_{f\sim\mathcal{F}}\left[\mathbb{E}_{\boldsymbol{A}\sim B(\boldsymbol{A})}\left[\frac{\partial \mathcal{L}_{ce}\left(f_\theta\left(\left(\boldsymbol{E}+\boldsymbol{\delta}'\right)\boldsymbol{A}\right), y\right)}{\partial \boldsymbol{\delta}}\right]\right] \\ &= \frac{\partial\left(\frac{1}{KB}\sum_{i=1}^{K}\sum_{j=1}^{B}\mathcal{L}_{ce}^{i,j}\right)}{\partial \boldsymbol{\delta}} = \frac{1}{KB}\sum_{i=1}^{K}\sum_{j=1}^{B}\frac{\partial \mathcal{L}_{ce}^{i,j}}{\partial \boldsymbol{\delta}} \end{aligned} \tag{16}$$

where B is the size of each mini-batch, $B(\boldsymbol{A})$ is the distribution of the abundance matrix corresponding to all patches of the mini-batch, and $\mathcal{L}_{ce}^{i,j} = \mathcal{L}_{ce}(f_\theta^i((\boldsymbol{E}+\boldsymbol{\delta}')\boldsymbol{A}_j), y_j)$.

It can be theoretically demonstrated that when small batches of patches are uniformly and randomly sampled from the training set, the expected gradient of the endmember matrix computed from these mini-batches is equal to the gradient computed over the entire training set. The expectation of the gradient estimated from mini-batch patches can be expressed as follows:

$$\begin{aligned} &\mathbb{E}\left[\frac{1}{KB}\sum_{i=1}^{K}\sum_{j=1}^{B}\frac{\partial \mathcal{L}_{ce}^{i,j}}{\partial \boldsymbol{\delta}}\right] = \mathbb{E}\left[\frac{1}{KB}\sum_{i=1}^{K}\sum_{j=1}^{N}\mathbb{I}(j)\frac{\partial \mathcal{L}_{ce}^{i,j}}{\partial \boldsymbol{\delta}}\right] \\ &= \frac{1}{KB}\sum_{i=1}^{K}\sum_{j=1}^{N}\mathbb{E}\left[\mathbb{I}(j)\right]\frac{\partial \mathcal{L}_{ce}^{i,j}}{\partial \boldsymbol{\delta}} = \frac{1}{KN}\sum_{i=1}^{K}\sum_{j=1}^{N}\frac{\partial \mathcal{L}_{ce}^{i,j}}{\partial \boldsymbol{\delta}} \end{aligned} \tag{17}$$

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Algorithm 2 Perturbation Update Computational Detail

Input: The mini-batch $[\boldsymbol{x}_t^1, \boldsymbol{x}_t^2, \ldots, \boldsymbol{x}_t^B]$; The endmember matrix $\boldsymbol{E}_t$; the hyperspectral classifiers $f_\theta^1, f_\theta^2, \ldots, f_\theta^K$; perturbation $\boldsymbol{\delta}_{t-1}$ after the $(t-1)$-th iteration; the size of perturbation ϵ; the step size α and the convergence hyperparameter c.
Output: Perturbation $\boldsymbol{\delta}_t$.

1: ***# Calculate Classification Losses*;**
2: Input $[\boldsymbol{x}_t^1, \boldsymbol{x}_t^2, \ldots, \boldsymbol{x}_t^B]$ to the classifiers $f_\theta^1, f_\theta^2, \ldots, f_\theta^K$.
3: Calculate the cross entropy loss $\mathcal{L}_{ce}^{i,j}$ $for\ i = 1, 2, \ldots K, j = 1, 2, \ldots, B$.
4: ***# Calculate Conflict-Mitigation Gradient*;**
5: Calculate the gradients $\boldsymbol{g}_i$ and $\boldsymbol{g}_0$ by

$$\boldsymbol{g}_i = \frac{1}{B}\sum_{j=1}^{B}\frac{\partial \mathcal{L}_{ce}^{i,j}}{\partial \boldsymbol{E}_t}, \quad \boldsymbol{g}_0 = \frac{1}{K}\sum_{i=1}^{K}\boldsymbol{g}_i. \tag{18}$$

6: Obtain the optimal w^* by solving Eq. (22).
7: Calculate the optimal λ^* and $\boldsymbol{\Delta}_t^*$ by Eq. (23).
8: ***# Update Adversarial Perturbation*;**
9: Update the adversarial perturbation $\boldsymbol{\delta}_t$ by Eq. (24).
10: **return** Adversarial perturbation $\boldsymbol{\delta}_t$.

where $\mathbb{I}(j)$ denotes an indicator function that equals 1 if the jth patch is included in the mini-batch and 0 otherwise. According to the aforementioned derivation, the gradient of the endmember matrix computed using the mini-batch is an unbiased estimate of the true gradient over the entire training set. This unbiasedness ensures that, over multiple iterations, the average of gradients computed from mini-batches converges to the true gradient of the full-training set, thereby guaranteeing that the direction of parameter updates is correct in expectation.

C. Conflict-Mitigation Gradient Optimization

During the optimization of the endmember matrix guided by backpropagated gradients from multiple hyperspectral classifiers, architectural differences among the classifiers (e.g., CNN-based and transformer-based models) may lead to conflicts in gradient directions. Such conflicts can weaken the attack effectiveness of the optimized adversarial endmember matrix [55], [56], [57]. To address this issue, EEGTAA introduces a conflict-mitigation gradient ascent strategy. Specifically, during the iterative optimization of the endmember matrix, the update vector $\boldsymbol{\Delta}_t \in \mathbb{R}^{L\times p}$ is required not only to increase the average classification loss across the ensemble but also to ensure that the loss for each individual classifier is elevated, thereby enhancing the adversarial effectiveness against diverse classifier architectures. For classifier f_θ^i, EEGTAA employs the $R(\boldsymbol{E}_t, \boldsymbol{\Delta}_t)$ to quantify the consistency of gradient directions among the integrated classifiers in the tth iteration, where $R(\boldsymbol{E}_t, \boldsymbol{\Delta}_t)$ is defined as follows:

$$\begin{aligned} &R\left(\boldsymbol{E}_t, \boldsymbol{\Delta}_t\right) \\ &= -\min_{i\in[1,\ldots,K]}\left\{\frac{1}{\alpha}\left(\frac{1}{B}\sum_{j=1}^{B}\mathcal{L}_{ce}\left(f_\theta^i\left(\left(\boldsymbol{E}_t+\alpha\boldsymbol{\Delta}_t\right)\boldsymbol{A}_j\right), y_j\right)\right.\right. \\ &\qquad \left.\left. -\frac{1}{B}\sum_{j=1}^{B}\mathcal{L}_{ce}\left(f_\theta^i\left(\boldsymbol{E}_t\boldsymbol{A}_j\right), y_j\right)\right)\right\} \\ &\approx -\min_{i\in[1,\ldots,K]}\langle \boldsymbol{g}_i, \boldsymbol{\Delta}_t\rangle \end{aligned} \tag{19}$$

where $\boldsymbol{g}_i = 1/B\sum_{j=1}^{B}\partial\mathcal{L}_{ce}^{i,j}/\partial\boldsymbol{E}_t$ denotes the endmember matrix gradient computed by the ith classifier. When the step size α is sufficiently small, the classification loss can be approximated using a first-order Taylor expansion. Under this approximation, if $R(\boldsymbol{E}_t, \boldsymbol{\Delta}_t) < 0$, it implies that the update direction $\boldsymbol{\Delta}_t$ leads to an increase in the loss for all classifiers under a small perturbation.

Inspired by the work of Liu et al. [56], we formulate the following optimization problem to derive an update vector $\boldsymbol{\Delta}_t$ that minimizes gradient conflicts among the ensemble models, thereby enhancing the effectiveness of adversarial perturbation optimization

$$\begin{aligned} &\max_{\boldsymbol{\Delta}_t\in\mathbb{R}^{L\times p}}\min_{i\in[1,\ldots,K]}\langle \boldsymbol{g}_i, \boldsymbol{\Delta}_t\rangle \\ &\text{s.t.}\|\boldsymbol{\Delta}_t - \boldsymbol{g}_0\| \le c\|\boldsymbol{g}_0\|_2 \end{aligned} \tag{20}$$

where $\boldsymbol{g}_0 = 1/KB\sum_{i=1}^{K}\sum_{j=1}^{B}\partial\mathcal{L}_{ce}^{i,j}/\partial\boldsymbol{E}_t$, $c \in [0, 1)$ is a hyperparameter controlling the convergence rate. However, the optimization problem described by (20) involves high-dimensional variable $\boldsymbol{\Delta}_t$, which poses significant computational challenges. To address this, it is common to reformulate the problem into its dual form, thereby reducing the optimization complexity. Specifically, first $\min_{i\in[1,\ldots,K]}\langle \boldsymbol{g}_i, \boldsymbol{\Delta}_t\rangle = \min_{w\in\mathcal{W}}\langle\sum_i^K w_i\boldsymbol{g}_i, \boldsymbol{\Delta}_t\rangle$, where $\mathcal{W} = \left\{w|\sum_i^K w_i = 1 \text{ and } w_i \ge 0\right\}$. The Lagrangian of the objective in (20) is

$$\min_{\lambda\ge0, w\in\mathcal{W}}\max_{\boldsymbol{\Delta}_t\in\mathbb{R}^{L\times p}} \boldsymbol{g}_w^T\boldsymbol{\Delta}_t - \lambda\frac{\|\boldsymbol{g}_0 - \boldsymbol{\Delta}_t\|^2}{2} + \lambda\frac{\phi}{2} \tag{21}$$

where $\boldsymbol{g}_w = \sum_i^K w_i\boldsymbol{g}_i$ and $\phi = c^2\|\boldsymbol{g}_0\|_2^2$. The optimal solution w^* is derived by solving the following equation:

$$w^* = \arg\min_{w\in\mathcal{W}} \boldsymbol{g}_w^T\boldsymbol{g}_0 + \sqrt{\phi}\|\boldsymbol{g}_w\|_2. \tag{22}$$

The optimal solutions for λ^* and $\boldsymbol{\Delta}^*$ can be obtained through the following closed-form expressions:

$$\lambda^* = \frac{\|\boldsymbol{g}_w^*\|_2}{\sqrt{\phi}}, \ \boldsymbol{\Delta}_t^* = \boldsymbol{g}_0 + \frac{\boldsymbol{g}_{w^*}}{\lambda^*}. \tag{23}$$

For detailed implementation and convergence analysis of the algorithm, please refer to [56]. Based on the obtained conflict-mitigation gradient $\boldsymbol{\Delta}_t^*$, the adversarial perturbation $\boldsymbol{\delta}_t$ is updated as follows:

$$\boldsymbol{\delta}_t = \text{CLIP}\left(\boldsymbol{\delta}_{t-1} + \alpha\frac{\boldsymbol{\Delta}_t^*}{\|\boldsymbol{\Delta}_t^*\|_2}, min = -\epsilon, max = \epsilon\right). \tag{24}$$

In summary, the optimization of the endmember matrix at iteration t involves three key steps: loss computation, conflict-mitigation gradient calculation, and perturbation update.

1) *Loss Computation:* According to $\boldsymbol{x}_t^i = \boldsymbol{E}_t\boldsymbol{A}_i$, the adversarial hyperspectral samples $[\boldsymbol{x}_t^1, \boldsymbol{x}_t^2, \ldots, \boldsymbol{x}_t^B]$ within the current mini-batch are generated. These samples are then fed into the classifier $[f_\theta^1, f_\theta^2, \ldots, f_\theta^k]$, and the corresponding classification losses

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$[(\mathcal{L}_{ce}^{1,1},\ldots,\mathcal{L}_{ce}^{1,K}),(\mathcal{L}_{ce}^{2,1},\ldots,\mathcal{L}_{ce}^{2,K}),\ldots,(\mathcal{L}_{ce}^{B,1}\ldots,\mathcal{L}_{ce}^{B,K})]$ are computed.

2) *Conflict-Mitigation Gradient Calculation:* According to $\boldsymbol{g}_i = 1/B\sum_{j=1}^{B}\partial\mathcal{L}_{ce}^{i,j}/\partial\boldsymbol{E}_t$, the endmember matrix gradient backpropagated from the jth classifier can be computed. The optimal weight vector w^* is then obtained by solving (22), which is subsequently substituted into (23) to derive the conflict-mitigation gradient $\boldsymbol{\Delta}_t^*$.

3) *Perturbation Update:* According to (24), the adversarial perturbation $\boldsymbol{\delta}_t$ on the endmember matrix is updated.

By performing the above computational steps at each iteration, EEGTAA effectively optimizes the adversarial perturbations applied to the endmember matrix. The optimized adversarial endmember matrix is then utilized to guide the generation of hyperspectral adversarial examples aimed at deceiving state-of-the-art hyperspectral classifiers. The detailed computation process at iteration t is presented in Algorithm 2.

D. Analysis

Existing adversarial attack methods for HSI classification often neglect the spectral reflectance consistency inherent in HSI data, leading to adversarial examples with low physical interpretability. To address this issue, this study proposes an endmember extraction guided adversarial attack framework named EEGTAA. By introducing adversarial perturbations directly on the endmember matrix, EEGTAA ensures that the generated adversarial examples retain clear physical meaning. To reduce the computational overhead during perturbation optimization, EEGTAA constructs mini-batches using spatially adjacent hyperspectral patches and computes the optimization gradients of the endmember matrix based on these mini-batches. Furthermore, considering the potential inconsistency in backpropagated gradient directions across classifiers with heterogeneous architectures, EEGTAA incorporates a conflict-mitigation gradient optimization strategy to mitigate the adverse effects of gradient conflicts on both optimization efficiency and adversarial effectiveness.

IV. Experimental Results

In the experimental section, we first introduce three publicly available HSI classification datasets along with the experimental settings. Subsequently, the proposed attack method is compared against existing approaches. Then, ablation studies are conducted to validate the effectiveness of the key components in the EEGTAA framework. Finally, a systematic analysis and discussion of hyperparameters are presented to further investigate their impact on attack performance.

A. Experimental Settings

1) Dataset: This study conducts experiments on three widely used public hyperspectral datasets, namely, the Indian Pines (IP) dataset, the Salinas (SA) dataset, and the Pavia University (PU) dataset. Detailed information about the three datasets is provided in Section I of the Supplementary Materials.

2) Classifier Networks: This study employs eleven well-trained HSI classifiers to evaluate the attack performance of the proposed method, that is, AlexNet [58], ResNet18 [59], ResNet34 [59], DenseNet121 (DenNet121) [60], DenseNet161 (DenNet161) [60], ViT-Base (ViT-B) [61], Swin Transformer-Tiny (Swin-T) [62], Swin Transformer-Small (Swin-S) [62], Swin Transformer v2-Tiny (Swin-v2-T) [63], Swin Transformer v2-Small (Swin-v2-S) [63], ReSC-Net [7], 3DAM-CAM DenseNet (3D-DNet) [64], and SSFBF [65]. Specifically, ReSC-Net introduces a network architecture that integrates attention-enhanced residual modules with spatial–channel attention modules. The 3D-DNet adaptively recalibrates channel features in DenNet to enhance cross-channel correlations, and SSFBF proposes a dual-path transformer architecture with dynamic routing fusion to improve the feature extraction capability for hyperspectral data. To ensure fairness, all classifiers share the same weights for each dataset in the adversarial attack comparison experiments.

3) Comparative Methods: In Section III, we compare the proposed method with eight adversarial attack algorithms, namely PGD [66], MIFGSM [23], VMIFGSM [24], SS-FGSM [29], I-DeepFool [28], SVRE [67], SMER [68], and AdaEA [69]. It is important to note that all methods perform adversarial attacks at the endmember level. Specifically, SS-FGSM applies perturbations to the endmember matrix in a single step, while the remaining methods generate adversarial examples iteratively. PGD uses random initialization within the constrained range. MIFGSM accumulates gradient information at each iteration to enhance the optimization stability, and VMIFGSM further incorporates gradient variance from the previous iteration to stabilize the update direction of perturbations. I-DeepFool searches along the shortest path from benign samples to the classification hyperplane, thereby generating adversarial examples with minimal perturbation. SVRE reduces the gradient variance of the ensemble models to prevent the optimization process from getting stuck in local optima. SMER leverages reinforcement learning to adaptively reweight ensemble models, thus improving the transferability of adversarial examples. Finally, AdaEA adaptively adjusts the fusion of model outputs by monitoring the differential contributions of each ensemble model to the adversarial objective, thereby enhancing attack effectiveness. A more detailed comparison between EEGTAA and the other baseline methods is provided in Section I of the Supplementary Materials.

4) Implementation Details: Before evaluating the performance of the adversarial attack, we first apply the VCA algorithm to perform the endmember extraction on the HSI to obtain the original endmember matrix and estimate the abundances. This information represents the physical properties inherent in the hyperspectral data itself and is independent of the classifier. In addition, to ensure the validity of the attack performance evaluation, all HSI classifiers are thoroughly pretrained. All HSI classifiers are trained for 100 epochs. The training set is constructed by randomly sampling 70% of the pixels from each class, with the remaining 30% of pixels used to form the testing set. Detailed configuration information of the classifiers is provided in Section I of the Supplementary

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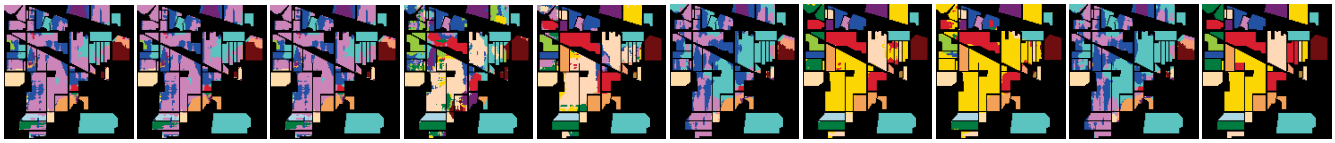


Fig. 3. Prediction results of ReSC-Net on the IP dataset under nine adversarial attack methods. (a) PGD. (b) MIFGSM. (c) VMIFGSM. (d) SS-FGSM. (e) I-DeepFool. (f) SVRE. (g) SMER. (h) AdaEA. (i) EEGTAA. (j) Ground truth.

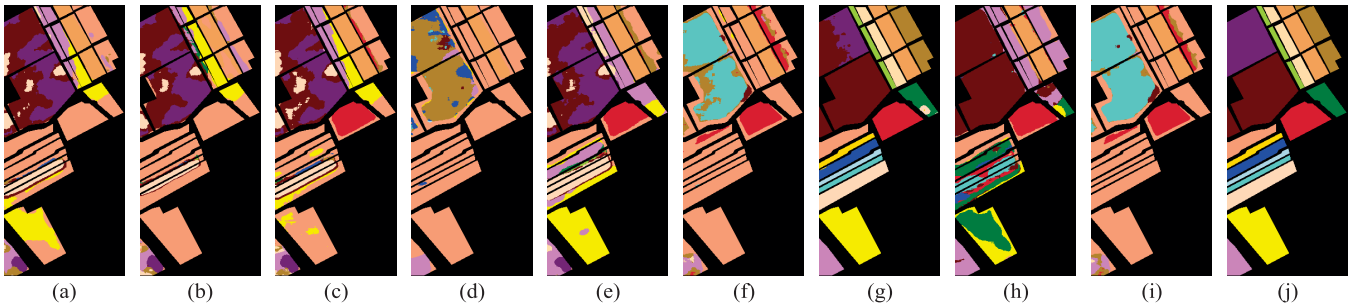


Fig. 4. Prediction results of ReSC-Net on the SA dataset under nine adversarial attack methods. (a) PGD. (b) MIFGSM. (c) VMIFGSM. (d) SS-FGSM. (e) I-DeepFool. (f) SVRE. (g) SMER. (h) AdaEA. (i) EEGTAA. (j) Ground truth.

Materials. To ensure fairness, only hyperspectral samples that are correctly classified by all classifiers are selected for testing the attack methods. The number of hyperspectral samples used for adversarial sample performance testing on the IP, SA, and PU datasets is 9501, 49 516, and 42 570, respectively. Compared with the complete set of labeled samples in each dataset, the hyperspectral samples used for testing in IP, SA, and PU exhibit mean shifts of 0.037, 0.050, and 0.002, respectively, and variance shifts of 0.022, 0.049, and 0.003, respectively. These distribution shifts are quantified using the L_2-norm distance of the mean vectors and covariance matrices. The size of the hyperspectral patch fed into the classifiers is set to 21, i.e., $H_p = W_p = 21$. The number of endmembers p extracted using the VCA method is set to 20. For parameter settings, all methods except SS-FGSM adopt the same configuration: the maximum perturbation magnitude is set to $\epsilon = 0.015$, the number of iterations $T_{max} = 200$, and the step size $\alpha = 0.00001$. For the SS-FGSM method, both the perturbation magnitude and the step size are set to 0.015. In the proposed method, the convergence hyperparameter is fixed at $c = 0.3$, and the mini-batch size is set to $B = 128$. Finally, the computational procedures for optimizing adversarial perturbations on the endmember matrix using other comparison methods are described in detail in Section I of the Supplementary Materials.

5) Evaluating Indicator: For evaluation metrics, the attack success rate (ASR), overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ) are employed to evaluate the attack effectiveness of the proposed method in comparison with other approaches. In general, lower values of OA, AA, and κ or a higher ASR indicate that the generated hyperspectral adversarial examples exhibit stronger attack capability. The detailed definitions and computation of these metrics can be found in [30]. In addition, we quantify the spectral distortion before and after the attack using the L_2-norm distance and the spectral angle mapper, denoted by SD and SAM, respectively. The two metrics are computed as follows:

$$\mathrm{SD} = \left(\frac{1}{NH_pW_pL} \sum_{k=1}^{N} \sum_{h=1}^{H_p} \sum_{w=1}^{W_p} \sum_{l=1}^{L} \left(\boldsymbol{x}_{adv,k}^{h,w,l} - \boldsymbol{x}_k^{h,w,l} \right)^2 \right)^{\frac{1}{2}} \tag{25}$$

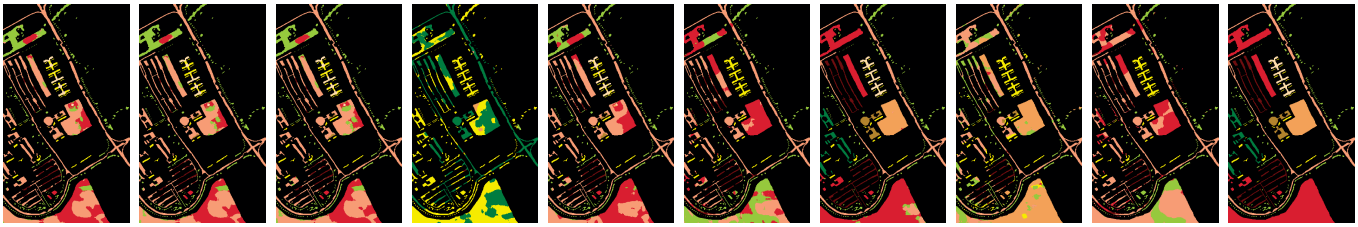
$$\mathrm{SAM} = \frac{1}{NH_pW_p} \sum_{k=1}^{N} \sum_{h=1}^{H_p} \sum_{w=1}^{W_p} \arccos \left(\frac{\left\langle \boldsymbol{x}_{adv,k}^{h,w}, \boldsymbol{x}_k^{h,w} \right\rangle}{\|\boldsymbol{x}_{adv,k}^{h,w}\| \cdot \|\boldsymbol{x}_k^{h,w}\|} \right) \tag{26}$$

where $\boldsymbol{x} \in \mathbb{R}^{L \times H_p \times W_p}$ and $\boldsymbol{x}_{adv} \in \mathbb{R}^{L \times H_p \times W_p}$ denote the clean hyperspectral patch and the adversarial hyperspectral patch generated under the guidance of the adversarial endmember matrix, respectively. N is the number of hyperspectral samples used for adversarial attack evaluation. $\langle \cdot, \cdot \rangle$ denotes the inner product, and $\| \cdot \|$ denotes the Euclidean norm. In general, smaller SD and SAM values indicate lower spectral distortion of the generated hyperspectral adversarial samples, and thus stronger stealthiness.

B. Attack Performance

In this section, we present a systematic comparison between the proposed EEGTAA method and several existing adversarial attack approaches. Specifically, ResNet18, DenNet121, Swin-T, and Swin-v2-T are integrated to optimize adversarial perturbations on the endmember matrix. The numerical results are summarized in Tables I–III, while the classification performance of ReSC-Net under adversarial attacks is illustrated in Figs. 3–5. It is important to note that the four white-box classifiers used to optimize the adversarial perturbations on the endmember matrix have access to detailed information, such as their internal parameters, gradients, and outputs. To better simulate a more realistic adversarial attack scenario, the remaining classifiers are kept in a black-box setup, where the

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 5. Prediction results of ReSC-Net on the PU dataset under nine adversarial attack methods. (a) PGD. (b) MIFGSM. (c) VMIFGSM. (d) SS-FGSM. (e) I-DeepFool. (f) SVRE. (g) SMER. (h) AdaEA. (i) EEGTAA. (j) Ground truth.

TABLE I

EVALUATION METRICS OF NINE ADVERSARIAL ATTACK METHODS ON ELEVEN PRETRAINED CLASSIFIERS OVER THE IP DATASET. THE ENSEMBLE CONSISTS OF RESNET18, DENNET121, SWIN-T, AND SWIN-V2-T, WITH THE MAXIMUM PERTURBATION MAGNITUDE SET TO $\epsilon = 0.015$. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Attack	Index	ResNet -18	ResNet -34	DenNet -121	DenNet -161	Swin -T	Swin -S	Swin -v2-T	Swin -v2-S	ReSC -Net	3D -DNet	SSFBF
PGD [66]	AA(%)↓	**6.25±1.56**	46.81±4.23	22.38±2.68	60.11±4.26	47.86±3.05	87.79±2.08	87.15±3.35	79.59±3.60	35.90±4.01	62.66±5.42	55.89±4.46
	OA(%)↓	9.48±0.95	35.15±3.58	18.13±1.83	45.31±4.52	39.95±3.92	77.46±2.34	72.37±2.73	66.58±3.41	34.08±3.47	51.44±4.82	49.39±4.93
	κ↓	**0.00±0.00**	0.32±0.09	0.15±0.06	0.42±0.13	0.35±0.10	0.75±0.11	0.70±0.11	0.63±0.13	0.31±0.09	0.48±0.15	0.45±0.15
	ASR(%)↑	90.51±0.92	64.56±3.52	80.41±1.87	53.14±4.49	60.04±3.98	22.46±2.33	27.62±2.71	32.76±3.46	65.78±3.44	47.67±4.78	50.60±4.96
MIFGSM [23]	AA(%)↓	6.25±0.65	47.12±3.48	22.27±2.95	60.18±4.72	50.31±4.38	88.57±2.25	88.06±2.94	81.11±3.14	34.17±3.35	63.66±4.79	56.48±4.71
	OA(%)↓	9.48±0.99	35.52±3.61	18.01±2.15	45.33±4.60	41.28±4.29	78.57±2.35	73.72±2.71	68.75±3.58	32.89±3.52	51.83±4.86	49.91±4.95
	κ↓	0.00±0.00	0.33±0.10	0.15±0.06	0.42±0.12	0.37±0.11	0.76±0.11	0.71±0.11	0.66±0.12	0.29±0.08	0.48±0.15	0.45±0.14
	ASR(%)↑	90.51±0.91	64.18±3.65	80.55±2.19	53.12±4.68	58.70±4.25	21.35±2.30	26.28±2.76	30.56±3.51	66.98±3.52	47.27±4.87	50.08±4.91
VMIFGSM [24]	AA(%)↓	6.25±1.48	46.96±4.12	21.68±2.52	60.74±4.20	50.35±4.16	88.67±2.19	88.02±2.78	80.78±3.31	34.34±3.57	64.12±4.27	56.56±4.80
	OA(%)↓	9.48±0.95	35.46±3.46	18.05±2.23	45.78±4.65	41.32±4.28	78.55±2.31	73.69±2.89	68.47±3.46	33.10±3.47	52.03±4.68	50.06±4.93
	κ↓	0.00±0.00	0.32±0.11	0.15±0.07	0.43±0.13	0.37±0.11	0.76±0.10	0.71±0.10	0.65±0.13	0.30±0.08	0.48±0.15	0.46±0.14
	ASR(%)↑	90.51±0.93	64.24±3.48	80.51±2.29	52.66±4.60	58.66±4.25	21.37±2.34	26.30±2.83	30.84±3.45	66.77±3.42	47.06±4.60	49.92±4.96
SS-FGSM [29]	AA(%)↓	6.25±1.23	**21.13±2.10**	44.61±3.50	57.04±4.68	54.32±4.47	88.75±2.51	84.59±3.36	**61.22±4.53**	56.76±4.78	79.64±3.52	66.33±4.25
	OA(%)↓	**3.77±0.41**	**23.80±2.26**	35.18±3.62	44.62±4.58	43.72±4.59	77.49±2.47	70.47±3.21	**53.82±4.67**	50.33±4.83	67.06±3.45	56.92±4.32
	κ↓	0.00±0.00	**0.19±0.08**	0.31±0.10	0.41±0.13	0.39±0.13	0.75±0.10	0.68±0.11	**0.50±0.15**	0.46±0.14	0.64±0.12	0.53±0.15
	ASR(%)↑	**96.23±0.48**	**76.00±2.20**	62.13±3.60	53.86±4.52	56.25±4.53	22.46±2.48	29.52±3.26	**45.65±4.67**	49.63±4.80	31.79±3.40	43.06±4.36
I-DeepFool [28]	AA(%)↓	59.02±4.73	83.96±3.15	58.16±4.36	87.57±2.96	95.63±1.30	98.79±0.52	99.84±0.24	96.78±0.39	85.33±4.12	88.45±2.90	96.28±1.18
	OA(%)↓	50.65±4.96	69.82±3.24	43.21±4.45	73.48±2.84	90.22±1.29	97.17±0.47	99.91±0.18	94.69±0.43	71.88±4.06	73.64±2.81	90.72±1.23
	κ↓	0.45±0.14	0.67±0.11	0.39±0.12	0.71±0.10	0.89±0.05	0.97±0.01	1.00±0.00	0.94±0.01	0.69±0.11	0.71±0.11	0.90±0.05
	ASR(%)↑	49.34±4.92	29.76±3.20	53.26±4.41	24.04±2.85	9.68±1.25	2.72±0.46	0.08±0.12	4.39±0.42	27.86±4.03	25.08±2.88	9.22±1.24
SVRE [67]	AA(%)↓	20.89±2.53	37.24±2.60	19.34±1.78	45.01±3.32	39.77±3.40	77.99±3.48	**83.34±3.39**	74.18±4.24	25.90±2.76	46.55±3.69	47.22±4.17
	OA(%)↓	26.54±2.60	28.76±2.72	17.22±1.64	33.46±3.40	34.68±3.39	68.27±3.39	**67.12±3.41**	60.82±4.10	27.31±2.68	38.81±3.70	41.16±4.27
	κ↓	0.19±0.08	0.26±0.09	0.15±0.07	0.30±0.09	0.30±0.09	0.65±0.10	**0.64±0.10**	0.57±0.13	0.22±0.10	0.35±0.11	0.36±0.11
	ASR(%)↑	73.46±2.68	71.00±2.76	81.41±1.68	65.42±3.48	65.33±3.30	31.67±3.37	**32.88±3.45**	38.58±4.13	72.61±2.66	60.52±3.74	58.84±4.25
SMER [68]	AA(%)↓	92.37±1.24	96.16±1.13	98.15±0.47	96.64±0.58	99.50±0.25	99.79±0.18	100.00±0.00	98.49±0.37	97.95±0.39	98.50±0.26	98.66±0.35
	OA(%)↓	92.16±1.06	93.30±0.97	98.36±0.36	95.06±0.68	99.56±0.14	99.78±0.09	99.99±0.00	98.38±0.22	97.01±0.43	98.40±0.28	99.81±0.19
	κ↓	0.91±0.05	0.92±0.05	0.98±0.01	0.94±0.02	0.99±0.00	1.00±0.00	1.00±0.00	0.98±0.01	0.97±0.01	0.98±0.01	1.00±0.00
	ASR(%)↑	7.83±1.02	5.95±0.90	0.33±0.34	2.29±0.61	0.21±0.17	0.07±0.08	0.00±0.01	0.62±0.24	2.89±0.41	0.66±0.23	0.11±0.14
AdaEA [69]	AA(%)↓	33.11±4.62	91.45±1.85	68.23±3.27	77.89±2.37	98.96±0.21	98.98±0.09	98.67±0.11	97.63±0.20	77.81±2.06	88.30±1.98	86.42±1.35
	OA(%)↓	47.03±4.80	86.76±1.52	70.72±3.18	74.48±2.57	99.08±0.14	99.03±0.16	98.17±0.29	97.14±0.33	80.63±2.31	82.12±1.68	89.74±0.97
	κ↓	0.34±0.11	0.85±0.06	0.65±0.12	0.70±0.10	0.99±0.00	0.99±0.01	0.98±0.01	0.97±0.02	0.77±0.11	0.79±0.11	0.88±0.05
	ASR(%)↑	52.96±4.82	12.50±1.58	25.90±3.10	23.14±2.58	0.69±0.17	0.83±0.14	1.81±0.31	2.29±0.38	19.08±2.35	16.48±1.62	10.21±0.95
EEGTAA	AA(%)↓	24.89±3.10	35.05±2.57	**18.94±1.55**	**43.09±3.49**	**39.54±3.41**	**76.42±3.48**	83.65±3.37	75.80±3.60	**24.25±2.70**	**44.68±3.95**	**45.08±4.34**
	OA(%)↓	30.66±3.24	26.94±2.40	**16.88±1.73**	**31.77±3.30**	**34.37±3.38**	**67.33±3.63**	67.63±3.48	62.98±3.76	**26.25±2.56**	**37.04±3.87**	**39.47±4.17**
	κ↓	0.24±0.09	0.24±0.09	**0.14±0.06**	**0.29±0.09**	**0.30±0.10**	**0.64±0.10**	0.65±0.10	0.59±0.09	**0.20±0.08**	**0.33±0.11**	**0.35±0.11**
	ASR(%)↑	69.35±3.20	72.84±2.44	**81.78±1.70**	**67.17±3.35**	**65.63±3.37**	**32.60±3.68**	32.37±3.47	36.39±3.72	**73.68±2.54**	**62.33±3.85**	**60.53±4.17**

attacker can only assess the success of the attack by querying the predicted class outputs of the hyperspectral adversarial samples.

The experimental results on the IP dataset (see Table I and Fig. 3) demonstrate that PGD, MIFGSM, and VMIFGSM achieve strong attack performance on ResNet18 and DenNet121. However, when hyperspectral adversarial samples generated by these methods are applied to other models, their effectiveness is significantly lower than that of the proposed method. This discrepancy arises because the gradients provided by different classifiers exhibit substantial directional inconsistency. The aforementioned methods simply optimize adversarial perturbations by equally averaging these gradients, without explicitly correcting for gradient conflicts. As a result, gradients from a few models may dominate the optimization, leading to perturbations that overfit the parameters of specific models. Consequently, the optimization process is prone to poor local optima, thereby reducing attack transferability to classifiers outside the ensemble.

Table II and Fig. 4 present the attack results on the SA dataset, where it is evident that the proposed method achieves superior attack performance on most classifiers. In contrast, the performance of I-DeepFool and SMER is markedly weaker than that of the other methods. This can be attributed to the fact that DeepFool-type methods are designed to compute the minimal perturbation required to push a sample across the decision boundary, primarily serving to evaluate model vulnerability under minimal perturbations. Their design objective is not to maximize ASRs or enhance cross-model transferability. For SMER, the method employs reinforcement learning to

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TABLE II

EVALUATION METRICS OF NINE ADVERSARIAL ATTACK METHODS ON 11 PRETRAINED CLASSIFIERS OVER THE SA DATASET. THE ENSEMBLE CONSISTS OF RESNET18, DENNET121, SWIN-T, AND SWIN-V2-T, WITH THE MAXIMUM PERTURBATION MAGNITUDE SET TO $\epsilon = 0.015$. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Attack	Index	ResNet -18	ResNet -34	DenNet -121	DenNet -161	Swin -T	Swin -S	Swin -v2-T	Swin -v2-S	ReSC -Net	3D -DNet	SSFBF
PGD [66]	AA(%)↓	6.26±0.49	35.57±3.60	45.21±4.18	26.69±3.32	16.78±1.59	21.11±2.27	23.94±3.09	67.18±3.77	25.32±3.40	50.61±4.15	37.42±4.38
	OA(%)↓	4.95±0.61	39.72±3.85	39.48±4.03	30.27±3.29	15.37±1.68	23.77±2.48	29.69±3.21	60.57±3.80	35.18±3.14	63.88±3.92	43.74±4.46
	κ↓	**0.00±0.00**	0.35±0.12	0.33±0.12	0.25±0.09	0.09±0.04	0.17±0.07	0.23±0.08	0.57±0.14	0.28±0.10	0.37±0.12	0.41±0.13
	ASR(%)↑	95.07±0.62	60.28±3.82	60.47±4.09	69.73±3.20	84.60±1.65	76.20±2.48	70.31±3.21	39.39±3.84	64.81±3.19	33.35±3.99	56.26±4.42
MIFGSM [23]	AA(%)↓	**6.25±0.57**	26.66±3.61	26.93±3.50	24.72±2.94	12.88±1.54	18.85±2.57	21.70±2.66	76.08±3.58	18.45±2.65	55.31±3.11	38.65±4.89
	OA(%)↓	4.95±0.62	34.02±3.56	35.92±3.47	28.61±3.08	11.35±1.33	23.78±2.48	26.67±2.78	67.29±3.44	24.09±2.66	68.00±3.04	45.87±4.60
	κ↓	0.00±0.00	0.28±0.10	0.29±0.10	0.23±0.09	0.05±0.02	0.16±0.04	0.18±0.06	0.64±0.12	0.16±0.06	**0.25±0.09**	0.56±0.14
	ASR(%)↑	95.07±0.65	65.98±3.50	63.59±3.48	71.39±3.06	88.63±1.35	76.18±2.41	73.32±2.79	32.64±3.42	75.91±2.67	29.55±3.08	54.11±4.62
VMIFGSM [24]	AA(%)↓	6.25±0.39	29.97±3.10	44.65±3.75	31.11±3.66	12.67±1.35	18.96±2.29	21.25±2.49	79.29±3.21	26.75±3.42	48.56±3.52	38.50±4.68
	OA(%)↓	4.95±0.52	36.59±3.45	40.75±3.83	34.24±3.57	11.05±1.22	23.71±2.18	26.09±2.78	70.53±3.06	33.72±3.39	67.18±3.41	46.48±4.72
	κ↓	0.00±0.00	0.30±0.10	0.34±0.11	0.29±0.11	0.05±0.02	0.16±0.06	0.17±0.06	0.68±0.11	0.27±0.10	0.43±0.13	0.54±0.14
	ASR(%)↑	95.07±0.53	63.41±3.40	58.83±3.85	65.76±3.59	88.93±1.28	76.25±2.14	73.91±2.74	29.39±3.08	66.28±3.37	30.88±3.48	53.49±4.76
SS-FGSM [29]	AA(%)↓	6.25±0.55	**13.26±2.39**	12.32±1.28	20.50±2.29	15.93±1.10	13.01±1.52	27.50±2.55	62.59±3.91	**15.54±1.34**	53.27±3.02	26.77±4.61
	OA(%)↓	**3.71±0.42**	18.27±2.08	10.75±1.14	17.16±2.03	9.58±1.21	12.20±1.34	27.07±2.66	63.27±3.82	**11.99±1.20**	68.02±3.10	43.93±4.58
	κ↓	0.00±0.00	0.11±0.04	**0.00±0.00**	0.09±0.05	0.06±0.02	0.07±0.03	0.22±0.08	0.59±0.13	**0.08±0.05**	0.41±0.13	0.56±0.14
	ASR(%)↑	**96.29±0.41**	81.73±2.06	90.19±1.13	82.84±2.05	90.40±1.20	87.79±1.39	72.93±2.64	36.60±3.82	**88.01±1.25**	28.75±3.16	56.09±4.56
I-DeepFool [28]	AA(%)↓	6.25±0.51	32.53±3.90	49.96±4.19	32.99±3.67	12.85±1.47	20.54±2.89	22.80±3.38	83.18±2.92	37.48±4.41	55.94±3.27	40.97±4.73
	OA(%)↓	4.95±0.63	37.88±3.88	41.56±4.20	36.31±3.42	11.28±1.42	24.78±2.58	27.26±3.01	74.24±2.69	48.06±4.82	70.84±2.91	48.59±4.90
	κ↓	0.00±0.00	0.31±0.10	0.35±0.12	0.31±0.11	0.05±0.01	0.17±0.07	0.19±0.07	0.72±0.10	0.42±0.13	0.54±0.14	0.42±0.13
	ASR(%)↑	95.07±0.65	62.12±3.82	58.06±4.26	63.69±3.40	88.70±1.49	75.18±2.54	72.74±3.02	25.70±2.68	51.94±4.88	30.24±2.97	51.38±4.93
SVRE [67]	AA(%)↓	17.17±3.42	17.41±1.72	9.44±1.21	15.51±1.36	13.01±0.48	11.40±0.47	16.30±1.34	41.73±3.56	19.48±2.35	49.81±4.08	18.93±2.64
	OA(%)↓	34.85±3.58	13.94±1.50	9.46±1.06	15.41±1.44	**5.00±0.39**	6.60±0.54	16.83±1.55	32.76±3.47	18.09±2.02	59.96±4.11	22.05±2.38
	κ↓	0.25±0.08	0.10±0.05	0.05±0.02	0.10±0.04	**0.02±0.01**	0.01±0.00	0.11±0.05	0.29±0.10	0.14±0.07	0.54±0.14	0.17±0.07
	ASR(%)↑	65.14±3.50	86.06±1.58	90.46±1.05	84.59±1.48	91.99±0.31	93.39±0.52	83.17±1.59	67.18±3.40	81.91±2.08	38.02±4.16	77.93±2.33
SMER [68]	AA(%)↓	93.63±0.42	70.90±2.32	99.17±0.12	81.78±0.24	99.90±0.11	99.79±0.07	99.99±0.02	99.65±0.15	95.67±1.34	91.29±1.64	99.89±0.13
	OA(%)↓	92.36±0.65	79.21±2.19	99.44±0.09	89.39±1.18	99.82±0.05	99.88±0.02	99.99±0.01	99.59±0.01	92.58±1.07	86.54±1.58	99.87±0.08
	κ↓	0.91±0.04	0.77±0.09	0.99±0.00	0.88±0.05	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00	0.92±0.03	0.85±0.06	1.00±0.00
	ASR(%)↑	7.64±0.63	20.79±2.17	0.26±0.05	10.61±1.10	0.09±0.02	0.11±0.06	0.01±0.01	0.23±0.01	7.41±1.05	8.03±1.54	0.06 ±0.07
AdaEA [69]	AA(%)↓	17.76±3.60	19.63±2.33	22.59±4.16	**11.18±1.24**	17.58±1.27	12.99±2.19	22.90±2.48	54.98±4.42	39.77±4.85	52.62±4.22	21.75±3.45
	OA(%)↓	36.50±3.80	23.98±2.41	41.80±4.32	**11.11±1.17**	11.17±1.30	17.02±2.09	23.29±2.34	57.95±4.28	48.12±4.92	**59.40±4.20**	29.03±3.06
	κ↓	0.26±0.09	0.18±0.08	0.33±0.10	**0.07±0.04**	0.07±0.03	0.11±0.04	0.17±0.07	0.53±0.14	0.42±0.13	0.54±0.14	0.21±0.08
	ASR(%)↑	63.49±3.82	76.02±2.47	57.62±4.30	**88.89±1.15**	88.81±1.39	82.96±2.01	76.70±2.37	41.94±4.24	51.88±4.91	**38.62±4.27**	70.95±3.08
EEGTAA	AA(%)↓	17.40±3.36	17.13±1.16	**7.93±1.03**	15.31±1.68	**12.13±1.24**	**6.62±0.49**	**16.26±1.91**	**41.13±3.48**	19.38±2.06	**46.53±4.12**	**18.51±2.24**
	OA(%)↓	35.40±3.52	**13.62±1.22**	**7.79±0.83**	14.24±1.56	7.14±0.96	**5.18±0.67**	**16.63±1.54**	**32.17±3.30**	18.38±1.99	59.48±3.94	**21.35±1.98**
	κ↓	0.25±0.09	**0.09±0.04**	0.04±0.01	0.09±0.03	0.04±0.03	**0.01±0.00**	**0.11±0.04**	**0.28±0.09**	0.15±0.06	0.53±0.14	**0.15±0.05**
	ASR(%)↑	64.59±3.57	**86.38±1.24**	**92.17±0.81**	85.76±1.50	**92.85±0.96**	**94.81±0.62**	**83.37±1.50**	**67.78±3.37**	81.62±1.92	38.52±3.94	**78.62±1.94**

dynamically assign weights to models within the ensemble at each iteration. We conjecture that such a nonuniform weighting strategy may not be well-suited for high-dimensional hyperspectral data: on the one hand, it may suppress loss gradients that are beneficial for improving perturbation transferability; on the other hand, the continuous updating of ensemble weights during iterations may cause conflicting perturbation directions between successive updates, thereby destabilizing the adversarial optimization and hindering convergence.

On the PU dataset, which exhibits higher spectral complexity, Table III and Fig. 5 demonstrate that the proposed EEGTAA method achieves the best attack performance across most classifiers. In contrast, AdaEA evaluates the directional discrepancies of adversarial gradients derived from different classifiers using cosine similarity, and directly zeroes out perturbations at positions with large discrepancies. This simplistic zeroing operation undermines the integrity of the perturbations and yields perturbations with smaller L_2-norms, thereby limiting their transferability to models outside the ensemble. Although SVRE also delivers strong attack performance on most classifiers, its double-loop optimization strategy for perturbations on the endmember matrix leads to significantly higher computational cost than EEGTAA. Moreover, observations of the OA and AA metrics indicate that EEGTAA consistently achieves strong and generalizable attack effectiveness across different land-cover classes. Finally, it is worth noting that all methods exhibit reduced attack performance against transformer-based classifiers, likely due to their more robust feature extraction and global modeling capabilities. Nevertheless, EEGTAA still outperforms competing methods on most transformer-based classifiers, highlighting its superior effectiveness. This result not only highlights the robustness and adaptability of the proposed approach when confronted with models of substantially different architectures but also further validates its advantage in enhancing the cross-architecture transferability of hyperspectral adversarial examples. These findings suggest that EEGTAA holds broader applicability and stronger threat potential in practical scenarios.

Finally, Section II of the Supplementary Materials evaluates the spectral distortion of hyperspectral adversarial samples generated by different methods on the three datasets. Overall, I-DeepFool typically produces adversarial samples with the smallest spectral distortion, but it achieves a relatively low ASR. In contrast, EEGTAA is able to generate highly threatening hyperspectral adversarial samples while keeping the spectral distortion reasonably well controlled.

C. Ablation Experiments

This section conducts comprehensive ablation experiments on three key components of the proposed method, namely

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TABLE III

EVALUATION METRICS OF NINE ADVERSARIAL ATTACK METHODS ON 11 PRETRAINED CLASSIFIERS OVER THE PU DATASET. THE ENSEMBLE CONSISTS OF RESNET18, DENNET121, SWIN-T, AND SWIN-V2-T, WITH THE MAXIMUM PERTURBATION MAGNITUDE SET TO $\epsilon = 0.015$. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Attack	Index	ResNet -18	ResNet -34	DenNet -121	DenNet -161	Swin -T	Swin -S	Swin -v2-T	Swin -v2-S	ReSC -Net	3D -DNet	SSFBF
PGD [66]	AA(%)↓	13.69±0.86	**67.56±3.87**	56.65±3.31	65.68±3.75	46.40±2.62	50.26±4.34	35.66±1.89	63.45±3.88	46.08±4.47	56.90±3.81	71.97±2.73
	OA(%)↓	4.89±0.61	**62.73±3.96**	31.77±3.25	40.59±3.86	26.61±2.73	42.50±4.12	18.48±2.03	38.85±3.94	44.30±4.58	38.06±3.76	74.95±2.60
	κ↓	0.01±0.00	0.52±0.14	0.25±0.09	0.33±0.10	0.21±0.08	0.33±0.10	0.13±0.05	0.32±0.11	0.32±0.10	0.28±0.09	0.67±0.10
	ASR(%)↑	95.11±0.65	**37.22±3.98**	68.23±3.24	59.41±3.84	73.39±2.71	57.52±4.11	81.51±2.06	61.13±3.93	55.65±4.52	61.94±3.79	25.04±2.67
MIFGSM [23]	AA(%)↓	13.48±1.12	68.40±3.59	56.73±3.31	65.57±4.33	45.97±2.58	50.05±4.32	35.50±2.19	62.97±3.87	45.76±4.46	56.91±3.47	71.79±2.29
	OA(%)↓	**4.81±0.69**	63.88±3.71	31.86±3.36	40.41±4.23	26.24±2.84	41.84±4.08	18.33±2.36	38.44±3.72	43.25±4.49	37.97±3.86	74.82±2.58
	κ↓	0.01±0.00	0.53±0.14	0.26±0.09	0.33±0.11	0.21±0.07	0.32±0.10	0.13±0.05	0.32±0.11	0.31±0.11	**0.27±0.10**	0.67±0.10
	ASR(%)↑	**95.19±0.68**	36.07±3.70	68.14±3.34	59.59±4.26	73.76±2.81	58.18±4.08	81.66±2.34	61.54±3.75	56.71±4.43	62.02±3.84	25.17±2.59
VMIFGSM [24]	AA(%)↓	13.64±0.74	69.52±3.72	57.48±3.17	65.76±3.70	45.42±2.89	49.54±4.36	35.29±2.15	62.49±4.07	46.16±4.36	56.78±3.79	71.59±2.17
	OA(%)↓	4.93±0.52	64.71±3.78	32.47±3.26	40.67±3.89	25.83±2.73	40.79±4.15	18.15±1.98	37.96±3.97	42.98±4.37	37.23±3.66	74.71±2.54
	κ↓	0.01±0.00	0.55±0.13	0.26±0.09	0.33±0.11	0.20±0.08	0.31±0.10	0.13±0.05	0.32±0.11	0.31±0.11	0.27±0.09	0.67±0.10
	ASR(%)↑	95.07±0.52	35.23±3.71	67.53±3.20	59.33±3.85	74.17±2.78	59.23±4.16	81.83±1.95	62.02±3.94	56.98±4.31	62.77±3.60	25.28±2.51
SS-FGSM [29]	AA(%)↓	**11.57±0.48**	70.70±3.28	58.29±2.74	70.93±4.41	67.46±4.83	69.35±4.61	74.45±4.23	89.40±2.38	58.88±1.19	55.82±4.47	86.73±1.45
	OA(%)↓	5.08±0.62	65.84±3.40	30.20±2.87	54.98±4.39	48.16±4.91	46.96±4.70	58.00±4.19	81.55±2.13	**5.79±0.48**	40.10±4.22	85.15±1.61
	κ↓	**0.00±0.00**	**0.51±0.15**	0.27±0.09	0.40±0.13	0.40±0.13	0.40±0.13	0.51±0.15	0.77±0.07	**0.02±0.01**	0.31±0.10	0.81±0.08
	ASR(%)↑	94.92±0.58	34.15±3.41	69.80±2.83	45.02±4.37	51.82±4.87	53.02±4.76	41.97±4.23	18.39±2.18	**94.20±0.51**	59.90±4.17	14.83±1.58
I-DeepFool [28]	AA(%)↓	16.69±0.86	71.78±3.41	59.94±3.78	68.56±4.82	53.96±3.08	59.63±4.19	41.72±2.84	72.83±4.48	51.39±4.89	63.21±4.37	78.16±1.91
	OA(%)↓	6.28±0.74	71.28±3.21	34.83±3.65	50.35±4.92	31.95±3.23	54.11±4.48	23.29±2.66	54.04±4.39	50.68±4.95	57.00±4.21	80.35±1.84
	κ↓	0.03±0.01	0.62±0.13	0.28±0.10	0.41±0.13	0.26±0.09	0.44±0.14	0.18±0.07	0.47±0.01	0.38±0.12	0.45±0.02	0.74±0.12
	ASR(%)↑	93.72±0.75	28.66±3.25	65.17±3.64	49.65±4.97	68.06±3.28	45.91±4.52	76.70±2.62	45.93±4.41	49.26±4.92	43.00±4.20	19.64±1.84
SVRE [67]	AA(%)↓	22.64±2.24	73.80±2.37	42.99±2.28	64.50±3.51	**36.32±1.78**	40.36±2.59	**28.44±1.37**	51.13±2.75	39.03±4.18	55.50±4.06	65.62±2.38
	OA(%)↓	19.94±2.04	79.17±2.26	22.42±2.11	38.14±3.75	**17.14±1.68**	27.19±2.64	**11.96±1.26**	30.08±2.89	39.71±4.11	37.17±3.87	74.26±2.69
	κ↓	0.06±0.03	0.71±0.08	0.16±0.05	0.32±0.11	**0.12±0.06**	0.18±0.07	**0.07±0.04**	0.25±0.09	0.25±0.08	0.29±0.10	0.65±0.12
	ASR(%)↑	80.06±2.08	20.76±2.21	77.58±2.18	61.86±3.79	**82.85±1.64**	72.84±2.60	**88.03±1.24**	69.90±2.87	60.25±4.16	62.83±3.83	25.74±2.73
SMER [68]	AA(%)↓	99.74±0.12	97.88±0.35	100.00±0.00	99.26±0.21	99.99±0.01	99.91±0.16	99.69±0.13	99.95±0.05	98.42±0.28	96.17±0.17	99.92±0.12
	OA(%)↓	99.85±0.10	93.51±0.79	100.00±0.00	99.41±0.23	99.99±0.02	99.77±0.18	99.31±0.09	99.93±0.04	95.91±0.56	84.96±1.86	99.96±0.23
	κ↓	1.00±0.00	0.92±0.04	1.00±0.00	0.99±0.00	1.00±0.00	1.00±0.00	0.99±0.01	1.00±0.00	0.95±0.02	0.81±0.06	1.00±0.00
	ASR(%)↑	0.15±0.12	6.41±0.71	0.00±0.00	0.59±0.21	0.01±0.04	0.22±0.15	0.63±0.07	0.03±0.05	4.00±0.57	15.04±1.89	0.03±0.22
AdaEA [69]	AA(%)↓	39.83±2.73	71.39±3.49	59.93±3.61	74.22±4.47	58.45±3.86	66.66±4.20	72.34±4.28	86.93±2.94	45.45±3.78	64.22±4.14	86.62±2.76
	OA(%)↓	25.01±2.46	67.71±3.45	34.20±3.49	44.67±4.40	35.06±3.67	40.07±4.17	45.32±4.33	66.29±3.40	35.90±3.68	39.98±4.15	74.61±2.46
	κ↓	0.15±0.06	0.51±1.00	0.28±0.10	0.40±0.13	0.27±0.09	0.33±0.11	0.39±0.12	0.60±0.12	0.28±0.09	0.33±0.11	0.69±0.10
	ASR(%)↑	74.99±2.47	32.26±3.46	65.80±3.54	55.33±4.44	64.93±3.68	59.92±4.13	54.65±4.29	33.70±3.48	64.07±3.62	60.01±4.17	25.37±2.49
EEGTAA	AA(%)↓	15.78±1.97	74.00±1.28	**40.34±2.19**	**64.13 ±3.94**	37.45±2.84	**39.16±2.48**	30.16±2.27	**50.88±3.10**	**34.15±3.19**	**54.92±3.73**	**64.12±2.58**
	OA(%)↓	17.12±1.88	79.89±1.21	**20.49±2.10**	**37.41±3.83**	18.35±1.69	**26.14±2.37**	13.24±1.49	**29.83±3.16**	29.90±3.24	**36.77±3.61**	**73.64±2.64**
	κ↓	0.02±0.01	0.72±0.10	**0.14±0.06**	**0.32±0.10**	0.13±0.06	**0.17±0.07**	0.08±0.04	**0.24±0.09**	0.14±0.07	0.29±0.11	**0.64±0.12**
	ASR(%)↑	82.88±1.87	20.04±1.28	**79.51±2.16**	**62.59±3.84**	81.64±1.71	**73.89±2.38**	86.75±1.53	**70.15±3.13**	70.07±3.20	**63.23±3.67**	**26.35±2.64**

VCA-based endmember extraction, ensemble-based adversarial attacks and conflict-mitigation gradient optimization. All experiments are independently executed 20 times, and the mean values are reported as the final results.

First, this section analyzes the sensitivity of EEGTAA to different endmember extraction methods. We replace the VCA endmember extraction method used in the proposed approach with N-FINDER [35], CLSUnSAL [49], and the deep autoencoder-based endmember extraction method (denoted by AEEE) [70]. The experimental results are reported in Section III of the Supplementary Materials. Overall, although the endmembers extracted by different methods differ numerically, as long as the endmember set can effectively linearly reconstruct the original image, the EEGTAA framework maintains effective attack performance. This observation strongly demonstrates that the core advantage of our method lies in the endmember-level perturbation paradigm itself, rather than relying on any specific extraction algorithm. VCA serves as the baseline in this article because it strikes a good balance between accuracy and efficiency and is one of the most widely used algorithms in hyperspectral remote sensing. In addition, we analyze the sensitivity of EEGTAA to the initial endmember matrix $\boldsymbol{E}$ obtained via the VCA method. Specifically, we test EEGTAA's robustness to imperfect initial endmember matrices by adding varying levels of noise to the original endmember matrix. Detailed experimental results and analysis are provided in Section III of the Supplementary Materials. The results indicate that the proposed method is robust to noise in the endmember matrix. When the noise magnitude is small, EEGTAA still generates hyperspectral adversarial samples with high threat potential, with only a slight decrease in ASR. However, when the uniform noise intensity exceeds 0.3, ASR exhibits a significant attenuation. This phenomenon is analyzed in detail in Section III of the Supplementary Materials.

Next, for the ensemble adversarial attack component, four classifiers are selected for integration, i.e., ResNet18 (A), DenNet121 (B), Swin-T (C), and Swin-v2-T (D), while ReSC-Net and SSFBF are used as target models. The experimental results on the three datasets are presented in Fig. 6. It can be observed that when adversarial examples are optimized using only a single HSI classifier as the surrogate model, the ASR against the target model is substantially reduced, rendering the threat less effective, even when the surrogate shares a similar architecture with the target. The underlying reason is that adversarial optimization on a single model tends to overfit the perturbation direction to the specific decision boundary and gradient information of that classifier, thereby producing locally optimal perturbations that are only effective for that model but fail to generalize to others with different internal structures. Moreover, the gradients provided by a single classifier may contain model-specific noise or

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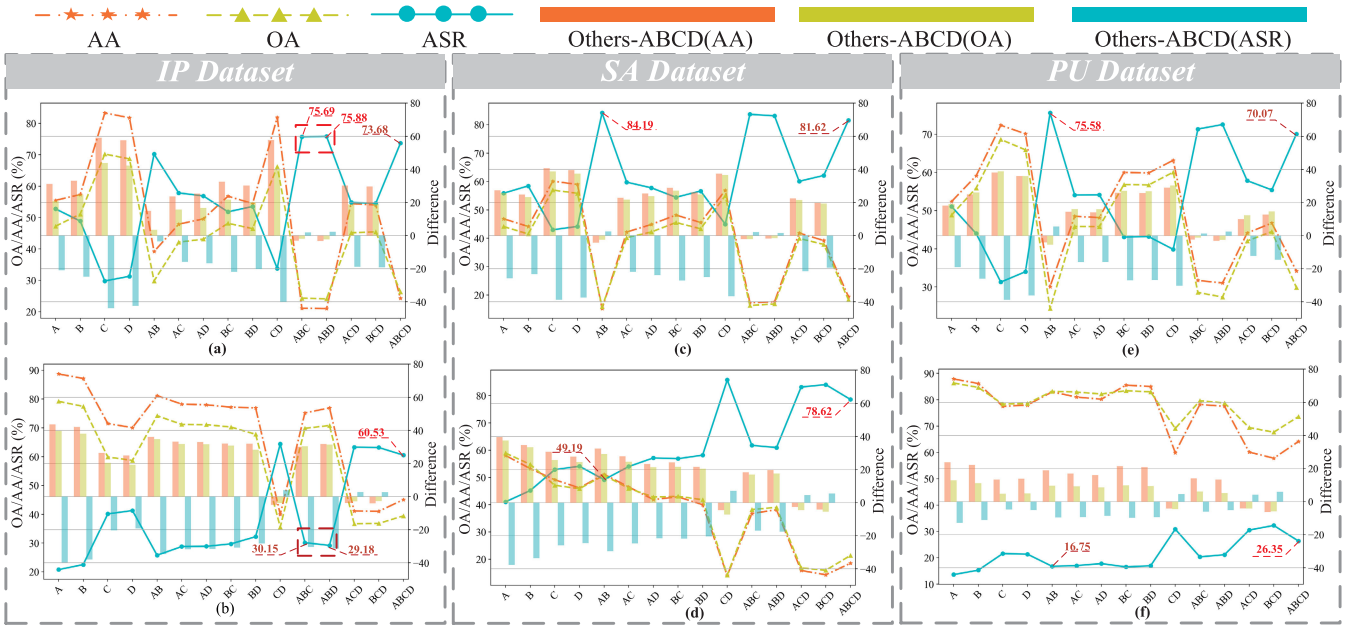


Fig. 6. Ablation results of the ensemble-based adversarial attack component on three datasets. The line plots report attack performance under different ensemble configurations, and the bar charts show the performance gap between each ensemble type and the four-model full-ensemble setting. (a), (c), and (e) Target model: ReSC-Net. (b), (d), and (f) Target model: SSFBF.

idiosyncratic information, making the resulting perturbation directions unstable and less representative.

Further observations indicate that on the IP dataset, adversarial examples generated by ensembles of ResNet18, DenNet121, and Swin-T (or ResNet18, DenNet121, and Swin-v2-T) often achieve higher ASRs against ReSC-Net than adversarial examples produced by an ensemble containing all four models (ResNet18, DenNet121, Swin-T, and Swin-v2-T). This phenomenon can be attributed to the high architectural similarity among ResNet18, DenNet121, and ReSC-Net, all of which are CNN-based classifiers; their decision-boundary patterns are therefore likely to be more similar. In contrast, Swin-T and Swin-v2-T are transformer-based and differ substantially in internal structure from the target model, and thus may contribute gradient signals that deviate from the optimal attack direction. A similar trend is observed on the SA and PU datasets: for the ReSC-Net target, adversarial examples generated by the ensemble of ResNet18 and DenNet121 exhibit the strongest threat. At the same time, comparison of Fig. 6(b), (d), and (f) shows that an ensemble configuration that is most effective against ReSC-Net yields substantially lower threat to SSFBF, which is transformer-based. This is likely because when the ensemble is dominated by CNN classifiers, the backpropagated gradients tend to favor perturbations that exploit vulnerabilities of CNNs and are therefore less effective against transformer models. Based on these findings, we reasonably infer that selecting an ensemble with a balanced mixture of architectures helps improve the expected attack success and transferability of generated hyperspectral adversarial examples against unknown target architectures. In addition, in Section III of the Supplementary Materials, we report the time and memory usage for different ensemble configurations. Overall, when the target model architecture is known, for example, a CNN or a transformer, ensembling two surrogate models with architectures similar to the target typically yields strong attack performance with a relatively modest computational overhead, while adding more models often provides only marginal gains. In contrast, when the target architecture is unknown, using a larger and more architecturally balanced ensemble of surrogate models usually leads to a higher expected attack performance, at the cost of substantially increased computational resources.

In addition, this section presents ablation studies on the conflict-mitigation gradient optimization strategy. Under unchanged settings for other hyperparameters, an ensemble of four HSI classifiers (ResNet18, DenNet121, Swin-T, and Swin-v2-T) is employed, and experiments are conducted on three datasets; the results are reported in Fig. 7. Fig. 7(b) visualizes the attack performance of this strategy on target classifiers, including ResNet34, DenNet161, Swin-S, Swin-v2-S, ReSC-Net, 3D-DNet, and SSFBF. Specifically, the version without the conflict-mitigation gradient optimization strategy is denoted as EEGTAA-N and is shown in blue in Fig. 7(b), while the version with the conflict-mitigation gradient optimization strategy is denoted as EEGTAA-Y and is shown in orange in Fig. 7(b). The results indicate that adversarial perturbations generated with the conflict-mitigation gradient optimization exhibit stronger attack potency and higher cross-model transferability across most HSI classifiers. This confirms that the proposed optimization mitigates performance degradation caused by directional inconsistencies among backpropagated gradients from classifiers with different architectures.

Furthermore, we perform a sensitivity analysis of the convergence hyperparameter c on ReSC-Net and SSFBF [see Fig. 7(a)]. When $c = 0$, EEGTAA degenerates to a standard

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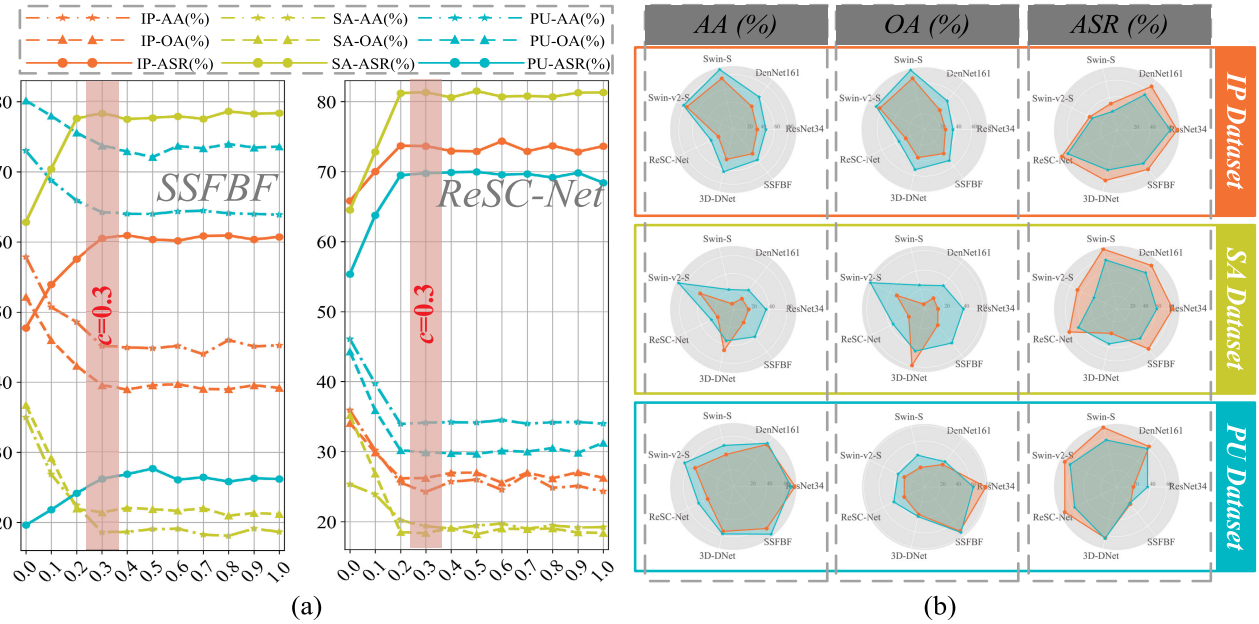


Fig. 7. Ablation results of the conflict-mitigation gradient optimization component. (a) Relationship between attack performance and the convergence hyperparameter c with target classifiers SSFBF and ReSC-Net. (b) Comparison of attack performance between EEGTAA-N (blue) and EEGTAA-Y (orange).

gradient-ascent attack and exhibits markedly reduced attack performance. As c increases, both attack strength and transferability of the generated adversarial examples improve. This indicates that the conflict-mitigation optimization effectively alleviates gradient conflicts across heterogeneous architectures and helps avoid convergence to poor local optima. Finally, we observe that when the convergence hyperparameter $c \geq 0.3$, the attack performance of the generated hyperspectral adversarial examples becomes stable across different target classifiers, with almost no significant fluctuations. This finding indicates that the proposed method exhibits low sensitivity to parameter variations within this range, thereby effectively avoiding performance degradation caused by minor parameter adjustments. In other words, the method demonstrates strong robustness and practicality in real-world applications, offering users a simple yet reliable hyperparameter selection strategy that further enhances the practical value of the proposed framework.

Finally, this section compares the robustness of basic adversarial defense strategies when perturbations are added to the endmember matrix versus when perturbations are added to the pixels. For the input hyperspectral adversarial samples, we applied spatial and spectral filtering with varying intensities. The results show that the hyperspectral adversarial samples generated by EEGTAA exhibit strong robustness to these basic defense strategies, whereas adversarial samples generated by adding perturbations to the pixels show weaker robustness. The experimental results and detailed analysis are recorded in Section III of the Supplementary Materials.

D. Parameter Analysis Experiments

This section first investigates the influence of two key parameters in EEGTAA, namely the step size α and the mini-batch size B. Under the condition that all other parameters remain fixed, the step size α is selected from the set $[1e^{-1}, 1e^{-2}, 1e^{-3}, 1e^{-4}, 1e^{-5}, 1e^{-6}]$, while the mini-batch size B is chosen from $[16, 32, 64, 128, 256, 512]$. In the experiments, ResNet18, DenNet121, Swin-T, and Swin-v2-T are integrated to generate adversarial perturbations, with ReSC-Net and SSFBF serving as target models. The corresponding experimental results and analysis are provided in Section IV of the Supplementary Materials. Overall, either excessively large or excessively small values of the iteration step size α and the mini-batch size B are detrimental to achieving optimal attack performance. Considering both computational efficiency and attack effectiveness, we recommend setting the step size to $\alpha = 1e^{-5}$ and the mini-batch size to $B = 128$.

In addition, this section further analyzes the influence of several key hyperparameters in EEGTAA on attack performance, including the perturbation magnitude ϵ, the number of iterations T_{max}, the patch size $H_p \times W_p$, and the number of endmembers p. The corresponding experimental results are reported in Section IV of the Supplementary Materials. Overall, increasing the endmember-level perturbation magnitude leads to improved attack effectiveness. However, excessively large perturbations tend to cause noticeable discrepancies between the spectra generated by the adversarial endmember matrix and the original clean spectra, thereby reducing stealthiness and introducing spectral distortion. With respect to the number of iterations T_{max}, the results indicate that the attack performance becomes stable with relatively small fluctuations when T_{max} exceeds 180. For the patch size and the number of endmembers, the attack performance exhibits an increasing-then-decreasing trend as these parameters increase. A more detailed analysis of these results is provided in Section IV of the Supplementary Materials.

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V. Concluding Remarks

This article proposes an EEGTAA method for HSI classification. To address the limitation of existing adversarial attack approaches that fail to ensure spectral reflectance consistency, EEGTAA first employs endmember extraction techniques to obtain the endmember composition of HSIs and introduces adversarial perturbations at the endmember level, thereby generating physically plausible adversarial examples. Subsequently, the method integrates the loss gradients of heterogeneous HSI classifiers to optimize perturbations on the endmember matrix, aiming to enhance the transferability of adversarial examples across classifiers with different architectures. Furthermore, EEGTAA incorporates a conflict-mitigation gradient optimization strategy during perturbation updating, which alleviates suboptimal optimization caused by large discrepancies in gradient directions across classifiers. In future work, we plan to further analyze the spectral characteristics of homogeneous hyperspectral pixels and develop adversarial perturbation optimization methods that are universally effective for pixels within the same class, thereby improving the overall efficiency of adversarial attacks on HSIs.

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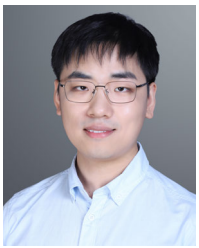
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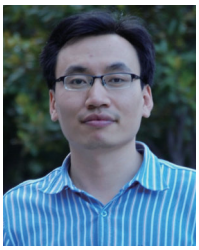
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